

Free-Fermion Trajectory Lyapunov Spectroscopy with Gain and Loss

Asadullah Bhuiyan

(Dated: August 6, 2026)

This note spells out how the trajectory-transfer construction used to extract operator scaling dimensions at measurement-induced transitions simplifies for number-conserving Gaussian fermion circuits. The many-body Kraus operator of a fixed measurement record is replaced, wherever possible, by one-body determinants, covariance updates, and singular values of one-body products. This simplification is not merely computational: it also exposes which “transfer matrix” is being diagonalized. We distinguish Born-sampled covariance trajectories, the trajectory-averaged mean channel, cumulative Gaussian/Kraus operator data, Choi particle and hole transfer charts, and the tangent cocycle used in finite-time Lyapunov simulations. Particular care is given to gain and loss. Single-fermion creation and annihilation branches are odd Gaussian amplitudes rather than ordinary invertible even Gaussian operators; exact projected branches therefore generate zero or infinite directions in one transfer chart. The correct finite diagnostic is the finite-sector particle or hole spectrum together with the null-sector occupation and multiplicity. With this bookkeeping in place, the usual cylinder formulas for the effective central charge, scaling dimensions, and multifractal moment exponents can be stated without conflating distinct operators.

CONTENTS

I. Introduction	2
II. Object Dictionary	2
III. Gaussian State Conventions	3
IV. Occupation Measurements as Determinantal Updates	3
V. Even Gaussian Branches: Normalization and One-Body Singular Values	4
A. One-Body Spectra Versus Fock-Space Spectra	5
VI. Gain and Loss: Odd Branches and Singular Transfer Charts	5
A. How Odd Branches Compose	6
B. Several Odd Insertions: Same-Type Blocks and Mixed Words	6
C. Transfer-Matrix Group: $O(2N, \mathbb{C})$ and the $U(1)$ Levi Chart	8
D. Moving Nambu Charts and the $O(2N, \mathbb{C})$ Regularization	9
E. Projective Measurements, Gauge Renormalization, and Particle/Hole Reciprocity	10
F. Majorana Transfer Form of Exact and Regularized Gain/Loss	12
G. Regulated Products in the Majorana/Nambu Chart	13
H. Laurent Limit of an Arbitrary Regulated Word	15
I. Case-by-Case Regulated Endpoint Limits	16
J. Dressed Projectors from the Single-Copy Nambu Regulator	20
K. Neutral Choi Reference Versus Odd-Operator Regularization	22
L. Operational Conclusion for Non-Commuting Gain/Loss Strings	25
VII. Tangent Cocycles for Covariance Dynamics	26
VIII. Trajectory Free Energy and CFT Finite-Size Data	27
A. Why Transfer Eigenvalues Are Free Energies	27
B. Replicas Are Not a Numerical Requirement	27
IX. Mechanical CPU Extraction Plan	29
X. Caveats and Non-Identifications	30
XI. Summary	31

I. INTRODUCTION

Zabalo *et al.* showed that a monitored circuit trajectory can be treated as a sample of a disordered statistical mechanics model by the identification

$$Z_{\xi} = p_{\xi}, \quad (1)$$

where ξ is the measurement record and p_{ξ} is its Born probability [1]. The logarithm of p_{ξ} is then a sample free energy, and the subleading singular values of the trajectory Kraus operator play the role of transfer-matrix eigenvalues. At a scale-invariant point, finite-size corrections of these Lyapunov data give the effective central charge and low-lying scaling dimensions of the corresponding non-unitary conformal field theory (CFT).

For an interacting random circuit, this is a many-body calculation. A trajectory operator acts on a Hilbert space of dimension exponential in system size, and its leading singular values must be extracted by a many-body iteration. In a number-conserving free-fermion circuit, the same conceptual construction has a much sharper form. A trajectory built from Gaussian operations remains Gaussian, so its probabilities are determinants and its finite transfer spectra are controlled by one-particle matrices or by covariance-matrix Schur complements. This is the sense in which the Zabalo–Gullans–Huse–Pixley spectroscopy becomes a free-fermion Lyapunov problem.

The purpose of this note is not to claim that every finite-time Lyapunov gap currently measured in the class-A adaptive circuit is already a universal CFT number. The purpose is to make the route precise enough that a numerical extraction cannot accidentally identify different objects. This matters especially in the adaptive circuit, where local feedback contains particle gain and loss. Those branches are perfectly legitimate Gaussian trajectory amplitudes, but they live on singular boundaries of the usual invertible particle-transfer chart.

The universal objects used below are standard but come from several literatures. Cylinder finite-size scaling and operator dimensions are reviewed in Refs. [2, 3]; entanglement central charge is a different use of conformal data [4]. Quenched Lyapunov spectra are governed by multiplicative ergodic theory [5, 6], while multifractal spectra at disordered critical points are reviewed in Ref. [7]. The free-fermion monitored-circuit context is developed in Refs. [8–12], and the topological Lyapunov/boundary-mode language relevant to class A appears in Refs. [13–15].

II. OBJECT DICTIONARY

The word “transfer” is overloaded. Table I fixes the objects used below.

TABLE I. Distinct objects that should not be identified without an explicit map.

Object	Definition	What it can diagnose
Born trajectory $G_{\xi}(t)$	State covariance evolved by the sampled record ξ	Trajectory-resolved nonlinear observables such as entropy, Chern markers, and correlators
Mean channel $\bar{G}(t)$	Deterministic covariance obtained after Born averaging the channel	Linear observables of the average state and mixed-state spectral skeletons
Cumulative Kraus/Gaussian operator	Unnormalized amplitude $\hat{K}_{\xi}(t)$ or its one-body representative when the branch is even and regular	Born probability, operator singular values, and trajectory free energy
Choi particle/hole transfer data	Doubled Gaussian covariance of $\hat{K}_{\xi}(t)$, represented in particle or hole charts plus endpoint data	Finite-sector singular values and singular gain/loss null sectors
Tangent cocycle	Linearization of $G_{t+1} = F_{\alpha_t}(G_t)$ along a realized trajectory	Stability exponents of covariance perturbations; near-marginal directions
CFT data	Universal finite-size coefficients of the trajectory ensemble at a critical point	c_{eff} , scaling dimensions, and multifractal spectra

The distinction between the third and fifth rows is essential. The cumulative Kraus or Choi transfer object asks for singular values of a realized operator. The tangent cocycle asks how an infinitesimal state perturbation evolves along the nonlinear stochastic covariance map. They may display the same slow physical sector, but they are not the same operator.

III. GAUSSIAN STATE CONVENTIONS

Let $c_i, c_i^\dagger$, $i = 1, \dots, N$, be complex fermions on the one-particle Hilbert space $\mathcal{H} \simeq \mathbb{C}^N$. For a Gaussian state ρ , define

$$C_{ij} = \text{Tr}(\rho c_i^\dagger c_j), \quad G = 2C - \mathbb{I}. \quad (2)$$

For a pure Slater determinant with occupied one-particle frame $Q \in \mathbb{C}^{N \times q}$, $Q^\dagger Q = \mathbb{I}_q$,

$$|\Psi_Q\rangle = \prod_{a=1}^q c^\dagger(Q_a)|0\rangle, \quad C = QQ^\dagger. \quad (3)$$

Hence

$$C^2 = C, \quad G^2 = (2C - \mathbb{I})^2 = \mathbb{I}. \quad (4)$$

Conversely, a Hermitian C with $C^2 = C$ is the one-body projector onto the occupied subspace of a fixed-charge pure Gaussian state. The covariance language is therefore equivalent to the Slater-frame language, but it is more convenient for measurements and feedback.

IV. OCCUPATION MEASUREMENTS AS DETERMINANTAL UPDATES

Let

$$|\chi\rangle \in \mathcal{H}, \quad P = |\chi\rangle\langle\chi|, \quad Q_\chi = \mathbb{I} - P, \quad \hat{n}_\chi = \hat{\chi}^\dagger \hat{\chi}, \quad (5)$$

with $\|\chi\| = 1$. The two projective measurement outcomes are $\hat{n}_\chi = 1$ and $\hat{n}_\chi = 0$. Their Born probabilities follow immediately from Eq. (2):

$$p_1 = \text{Tr}(PC) = \langle\chi|C|\chi\rangle, \quad p_0 = 1 - p_1 = \text{Tr}[P(\mathbb{I} - C)]. \quad (6)$$

The conditional covariance can be derived by choosing a basis in which χ is the first orbital and writing

$$C = \begin{pmatrix} c & v^\dagger \\ v & R \end{pmatrix}, \quad c = \langle\chi|C|\chi\rangle. \quad (7)$$

For the occupied outcome, Wick contraction conditioned on $n_\chi = 1$ gives

$$C_{\text{rest}}^{(1)} = R - \frac{vv^\dagger}{c}. \quad (8)$$

For the empty outcome, the determinantal complement gives

$$C_{\text{rest}}^{(0)} = R + \frac{vv^\dagger}{1 - c}. \quad (9)$$

Restoring basis-independent notation,

$$C^{(1)} = P + Q_\chi C Q_\chi - \frac{Q_\chi C P C Q_\chi}{\text{Tr}(PC)}, \quad (10)$$

$$C^{(0)} = Q_\chi C Q_\chi + \frac{Q_\chi C P C Q_\chi}{1 - \text{Tr}(PC)}. \quad (11)$$

The cross blocks with the measured mode vanish because the measured mode is now a classical definite occupation.

For several commuting measured modes with projector P_S , the same calculation gives determinant minors. If S denotes the set of modes conditioned to be occupied, then

$$\mathbb{P}(S \text{ occupied}) = \det(C_S), \quad (12)$$

where $C_S = P_S C P_S$ restricted to $\text{Ran } P_S$. Empty conditions are obtained by replacing C with $\mathbb{I} - C$, or by using the usual determinantal point-process conditional kernel. This is the first free-fermion simplification of $Z_\xi = p_\xi$: the trajectory weight is assembled from determinant factors rather than from a many-body tensor contraction.

V. EVEN GAUSSIAN BRANCHES: NORMALIZATION AND ONE-BODY SINGULAR VALUES

Consider an even number-conserving Gaussian branch whose action on creation operators is represented by a one-body matrix A :

$$\hat{K}_A c^\dagger(\phi) \hat{K}_A^{-1} = c^\dagger(A\phi), \quad (13)$$

with A invertible for the moment. Acting on a Gaussian density matrix with covariance C , the unnormalized output $\hat{K}_A \rho \hat{K}_A^\dagger$ has normalization

$$p_A(C) = \text{Tr}(\hat{K}_A \rho \hat{K}_A^\dagger) = \det(\mathbb{I} - C + ACA^\dagger), \quad (14)$$

up to the scalar prefactor used to normalize $\hat{K}_A$. The normalized output covariance is

$$C' = ACA^\dagger (\mathbb{I} - C + ACA^\dagger)^{-1}. \quad (15)$$

Thus the single-particle version of the many-body trace $\text{Tr}(\hat{K}_A \rho \hat{K}_A^\dagger)$ is not a one-body trace. It is the determinant in Eq. (14). In the $G = 2C - \mathbb{I}$ convention this same normalization is

$$p_A(G) = \det \left[\frac{\mathbb{I} - G}{2} + A \frac{\mathbb{I} + G}{2} A^\dagger \right]. \quad (16)$$

With the convention of Eq. (13), the mixed-state formula is $\det(\mathbb{I} - C + ACA^\dagger)$. One should not replace it by $\det(\mathbb{I} - C + CA^\dagger A)$ in a mixed state unless an additional convention or commutation relation justifies the reordering. The $A^\dagger A$ form appears cleanly after restricting to the occupied Slater frame, as in Eq. (18) below. Equation (15) is fixed by two checks. First, if A is unitary, it reduces to $C' = ACA^\dagger$. Second, in a basis where both C and $A^\dagger A$ are diagonal, each occupation mode is reweighted by the ordinary Bernoulli factor

$$n'_a = \frac{s_a^2 n_a}{1 - n_a + s_a^2 n_a}, \quad (17)$$

where s_a is the corresponding singular value of A .

For a pure Slater determinant, Eq. (3) gives the simpler Gram-determinant formula

$$p_A(Q) = \det(Q^\dagger A^\dagger A Q), \quad C' = A Q (Q^\dagger A^\dagger A Q)^{-1} Q^\dagger A^\dagger, \quad (18)$$

provided AQ has rank q . This formula is just the norm and projector of the transported occupied frame.

For a trajectory $\xi = (\alpha_1, \dots, \alpha_t)$ made only of regular even branches, the one-body product is

$$A_\xi(t) = A_{\alpha_t} \cdots A_{\alpha_1}. \quad (19)$$

If

$$s_a[A_\xi(t)] \sim \exp(t\gamma_a), \quad (20)$$

then the many-body Fock-space singular values of the second-quantized operator are products of the one-body singular values. Indeed, singular-value decomposing $A_\xi = U \text{diag}(s_1, \dots, s_N) V^\dagger$, the second-quantized unitaries $\Gamma(U)$ and $\Gamma(V)$ do not change singular values, while $\Gamma(\text{diag } s)$ acts on a Fock basis state $|I\rangle$ as

$$\Gamma(\text{diag } s)|I\rangle = \left(\prod_{a \in I} s_a \right) |I\rangle. \quad (21)$$

Thus

$$\lambda_I = \lim_{t \rightarrow \infty} \frac{1}{t} \log \sigma_I^2 = 2 \sum_{a \in I} \gamma_a, \quad (22)$$

apart from any trajectory-independent scalar normalization convention. In a fixed-charge sector, the allowed sets I have fixed cardinality.

A. One-Body Spectra Versus Fock-Space Spectra

It is useful to state the spectral dictionary without CFT language. Let $A_{\xi}(t)$ be any finite one-body Gaussian transfer matrix associated with a fixed trajectory and a fixed particle/hole chart. Write its singular values as

$$s_1(t) \geq s_2(t) \geq \cdots \geq s_N(t) > 0, \quad \kappa_a = \lim_{t \rightarrow \infty} \frac{1}{t} \log s_a(t). \quad (23)$$

The induced operator on the q -particle sector is the exterior power $\bigwedge^q A_{\xi}(t)$. Its singular values are

$$s_I^{(q)}(t) = \prod_{a \in I} s_a(t), \quad I \subset \{1, \dots, N\}, \quad |I| = q, \quad (24)$$

and its Lyapunov exponents are

$$\kappa_I^{(q)} = \sum_{a \in I} \kappa_a. \quad (25)$$

Thus the many-body spectrum is not a new diagonalization problem once the one-body singular data are known, but neither is it identical to the one-body list. It is the additive spectrum of the appropriate exterior-power or Fock-space representation.

The universal gaps are gaps relative to the dominant many-body configuration in the sector being measured. If I_0 maximizes $\kappa_I^{(q)}$ in the chosen q -particle sector, then

$$\mu_{I \leftarrow I_0}^{(q)} = \kappa_{I_0}^{(q)} - \kappa_I^{(q)} = \sum_{a \in I_0} \kappa_a - \sum_{a \in I} \kappa_a \geq 0. \quad (26)$$

For example, a single particle-hole replacement $a \in I_0 \rightarrow b \notin I_0$ has

$$\mu_{a \rightarrow b}^{(q)} = \kappa_a - \kappa_b. \quad (27)$$

Only in a convention where the low-energy one-body data have already been converted to positive excitation rapidities ϵ_ν measured relative to I_0 does the many-body tower take the simple form

$$\mu_A = \sum_{\nu} n_{\nu}(A) \epsilon_{\nu}. \quad (28)$$

This is the convention used by the finite-gap estimator in the CPU analysis script. The integer pattern $n_{\nu}(A)$ is not fixed by linear algebra alone: it is fixed by the charge, parity, particle/hole chart, boundary condition, and observable sector. The default choice “add the lowest one positive rapidity” is therefore a sector assumption, not a theorem.

If the trajectory transfer carries a scalar normalization $c_{\xi}(t)\Gamma(A_{\xi}(t))$, then $t^{-1} \log |c_{\xi}(t)|$ shifts every many-body exponent in the sector by the same amount. It affects absolute free-energy densities but cancels from the gaps $\mu_{I \leftarrow I_0}^{(q)}$. This is why CFT dimensions are extracted from many-body gaps, while c_{eff} is extracted separately from the trajectory probability free energy.

VI. GAIN AND LOSS: ODD BRANCHES AND SINGULAR TRANSFER CHARTS

The adaptive class-A circuit also uses feedback that supplies or removes a fermion from a measured mode. These operations are not ordinary even invertible Gaussian branches. On a pure Slater state they are

$$\hat{K}_+ = \hat{\chi}^\dagger, \quad \hat{K}_- = \hat{\chi}, \quad (29)$$

possibly composed with surrounding even Gaussian evolution. If the branch has nonzero amplitude, creation and annihilation update the physical covariance by

$$C_{\text{add}} = C + \frac{(\mathbb{I} - C)P(\mathbb{I} - C)}{\text{Tr}[(\mathbb{I} - C)P]}, \quad (30)$$

$$C_{\text{rem}} = C - \frac{CPC}{\text{Tr}(CP)}. \quad (31)$$

Equation (30) says that creation fills the component of χ that was empty; Eq. (31) says that annihilation removes the component of χ that was occupied. The denominators are the corresponding branch probabilities.

$$p_+(C) = \text{Tr}[P(\mathbb{I} - C)] = 1 - \langle \chi | C | \chi \rangle, \quad p_-(C) = \text{Tr}(PC) = \langle \chi | C | \chi \rangle. \quad (32)$$

In the $G = 2C - \mathbb{I}$ convention these become

$$p_+(G) = \frac{1 - \langle \chi | G | \chi \rangle}{2}, \quad p_-(G) = \frac{1 + \langle \chi | G | \chi \rangle}{2}. \quad (33)$$

A. How Odd Branches Compose

The important point for transfer matrices is that gain and loss do compose with the transfer product, but not as ordinary multiplication by another invertible one-body matrix in the same charge sector. Let $\mathcal{H}_1$ be the one-particle Hilbert space and let $\Gamma(A)$ denote the second-quantized even Gaussian operator induced by a one-body map A . On the q -particle sector $\bigwedge^q \mathcal{H}_1$, define exterior multiplication and contraction by

$$\varepsilon_\chi \Psi = \chi \wedge \Psi, \quad \varepsilon_\chi : \bigwedge^q \mathcal{H}_1 \rightarrow \bigwedge^{q+1} \mathcal{H}_1, \quad (34)$$

$$\iota_\chi(v_1 \wedge \cdots \wedge v_q) = \sum_{a=1}^q (-1)^{a-1} \langle \chi | v_a \rangle v_1 \wedge \cdots \widehat{v}_a \cdots \wedge v_q, \quad \iota_\chi : \bigwedge^q \mathcal{H}_1 \rightarrow \bigwedge^{q-1} \mathcal{H}_1. \quad (35)$$

Here $\widehat{v}_a$ means that v_a is omitted. The odd Kraus branches are precisely these charge-changing maps, up to the scalar normalization and fermion-ordering convention:

$$\hat{\chi}^\dagger \leftrightarrow \varepsilon_\chi, \quad \hat{\chi} \leftrightarrow \iota_\chi. \quad (36)$$

They obey the exact composition identities

$$\Gamma(A) \varepsilon_\chi = \varepsilon_{A\chi} \Gamma(A), \quad (37)$$

$$\iota_\chi \Gamma(A) = \Gamma(A) \iota_{A^\dagger \chi}. \quad (38)$$

Thus, for two even segments A_1, A_2 , a gain insertion between them gives

$$\Gamma(A_2) \varepsilon_\chi \Gamma(A_1) = \varepsilon_{A_2 \chi} \Gamma(A_2 A_1), \quad (39)$$

while a loss insertion gives

$$\Gamma(A_2) \iota_\chi \Gamma(A_1) = \Gamma(A_2 A_1) \iota_{A_1^\dagger \chi}. \quad (40)$$

Equations (39) and (40) are the mechanical reason gain/loss cannot be represented as a harmless extra factor in the same one-body transfer matrix product. The trajectory transfer is a word in even maps and charge-changing exterior/interior maps,

$$\widehat{\mathcal{T}}_\xi = \Gamma(A_m) \mathcal{O}_m \Gamma(A_{m-1}) \mathcal{O}_{m-1} \cdots \mathcal{O}_1 \Gamma(A_0), \quad \mathcal{O}_j \in \{\varepsilon_{\chi_j}, \iota_{\chi_j}\}, \quad (41)$$

so its many-body singular spectrum is a charge-sector-changing Fock-space spectrum. If one insists on a fixed particle-transfer chart, each ε_χ or ι_χ forces a rank-one direction to an endpoint or to a null direction, and the remaining finite block is the reduced transfer matrix used below. The missing endpoint occupation is not lost information; it is the record of which exterior/interior map occurred.

B. Several Odd Insertions: Same-Type Blocks and Mixed Words

The single-insertion discussion is enough to identify the obstruction, but it is not enough for an actual adaptive trajectory. A trajectory can contain several gains, several losses, or a mixed ordered string of both. The correct rule is not to commute the odd events into a convenient order. The correct rule is to keep the chronological exterior-algebra word.

First consider a block containing only gains. For

$$\widehat{\mathcal{T}}_+ = \Gamma(A_r)\varepsilon_{\chi_r}\Gamma(A_{r-1})\varepsilon_{\chi_{r-1}}\cdots\varepsilon_{\chi_1}\Gamma(A_0), \quad (42)$$

successive use of Eq. (37) gives

$$\widehat{\mathcal{T}}_+ = \varepsilon_{\eta_r}\varepsilon_{\eta_{r-1}}\cdots\varepsilon_{\eta_1}\Gamma(A_{\text{tot}}), \quad A_{\text{tot}} = A_r A_{r-1} \cdots A_0, \quad (43)$$

with transported gain modes

$$\eta_j = A_r A_{r-1} \cdots A_j \chi_j. \quad (44)$$

Thus the q -particle action is

$$\Psi \in \bigwedge^q \mathcal{H}_1 \quad \mapsto \quad \eta_r \wedge \eta_{r-1} \wedge \cdots \wedge \eta_1 \wedge A_{\text{tot}}^{\wedge q} \Psi \in \bigwedge^{q+r} \mathcal{H}_1. \quad (45)$$

Permuting two gain insertions changes the sign of the exterior product, and with interleaved even maps it also changes the transported vectors. Therefore only a block of adjacent gains in the same chart can be regarded as order-independent up to an overall sign.

For adjacent gains acting on a state with covariance C , put the measured modes in the matrix

$$X = (\chi_1, \dots, \chi_r), \quad E_X = X^\dagger (\mathbb{I} - C) X. \quad (46)$$

Wick's theorem gives the Born probability

$$p_{+X}(C) = \text{Tr} \left[\hat{\chi}_r^\dagger \cdots \hat{\chi}_1^\dagger \rho \hat{\chi}_1 \cdots \hat{\chi}_r \right] = \det E_X. \quad (47)$$

When E_X is nonsingular, the conditional covariance is

$$C_{+X} = C + (\mathbb{I} - C) X E_X^{-1} X^\dagger (\mathbb{I} - C). \quad (48)$$

This is the multi-mode version of Eq. (30): the branch fills the r -dimensional component of the requested gain subspace that lies in the empty subspace of the incoming Gaussian state. If $\det E_X = 0$, the branch is Pauli blocked in at least one linear combination; the exact trajectory weight is zero and the corresponding transfer chart is rank deficient.

The loss-only block is dual. For

$$\widehat{\mathcal{T}}_- = \Gamma(A_r)\iota_{\chi_r}\Gamma(A_{r-1})\iota_{\chi_{r-1}}\cdots\iota_{\chi_1}\Gamma(A_0), \quad (49)$$

successive use of Eq. (38) gives

$$\widehat{\mathcal{T}}_- = \Gamma(A_{\text{tot}})\iota_{\zeta_r}\iota_{\zeta_{r-1}}\cdots\iota_{\zeta_1}, \quad (50)$$

where

$$\zeta_j = (A_{j-1}A_{j-2}\cdots A_0)^\dagger \chi_j. \quad (51)$$

For adjacent losses in the current state chart,

$$O_X = X^\dagger C X, \quad (52)$$

and

$$p_{-X}(C) = \text{Tr} \left[\hat{\chi}_r \cdots \hat{\chi}_1 \rho \hat{\chi}_1^\dagger \cdots \hat{\chi}_r^\dagger \right] = \det O_X, \quad (53)$$

$$C_{-X} = C - C X O_X^{-1} X^\dagger C. \quad (54)$$

The determinant is now the occupied-subspace Gram determinant. A zero determinant means that at least one requested removal direction has no occupied component left after the previous removals.

The genuinely delicate case is a mixed string. Let

$$d_j = \begin{cases} \hat{\chi}_j^\dagger, & \sigma_j = +, \\ \hat{\chi}_j, & \sigma_j = -, \end{cases} \quad \hat{K}_\sigma = d_r d_{r-1} \cdots d_1. \quad (55)$$

The exterior and interior maps satisfy the Clifford relations

$$\{\varepsilon_u, \varepsilon_v\} = 0, \quad \{\iota_u, \iota_v\} = 0, \quad \iota_u \varepsilon_v + \varepsilon_v \iota_u = \langle u|v \rangle. \quad (56)$$

The last relation is the whole non-commuting issue. Moving a loss past a gain does not merely generate a sign; it generates a scalar contraction. In the simplest same-mode example,

$$\hat{\chi} \hat{\chi}^\dagger = 1 - \hat{n}_\chi, \quad \hat{\chi}^\dagger \hat{\chi} = \hat{n}_\chi. \quad (57)$$

A gain followed by a loss projects onto the initially empty sector, whereas a loss followed by a gain projects onto the initially occupied sector. These two branches have probabilities $1 - n_\chi$ and n_χ , respectively; they are not two equivalent endpoint insertions.

For an arbitrary mixed ordered string the compact formula is a Wick/Pfaffian formula. Define the ordered list of linear fermion operators appearing in $\hat{K}_\sigma^\dagger \hat{K}_\sigma$,

$$L_1, \dots, L_{2r} = d_1^\dagger, \dots, d_r^\dagger, d_r, \dots, d_1. \quad (58)$$

For an even gauge-invariant Gaussian state, the only nonzero elementary contractions are

$$\langle \hat{c}^\dagger(u) \hat{c}(v) \rangle = \langle u|C|v \rangle, \quad \langle \hat{c}(u) \hat{c}^\dagger(v) \rangle = \langle u|(\mathbb{I} - C)|v \rangle. \quad (59)$$

Let W be the $2r \times 2r$ antisymmetric matrix

$$W_{\mu\nu} = \begin{cases} \langle L_\mu L_\nu \rangle, & \mu < \nu, \\ -\langle L_\nu L_\mu \rangle, & \mu > \nu, \\ 0, & \mu = \nu. \end{cases} \quad (60)$$

Then

$$p_\sigma(C) = \text{Tr}(\hat{K}_\sigma \rho \hat{K}_\sigma^\dagger) = \langle L_1 \cdots L_{2r} \rangle = \text{Pf } W. \quad (61)$$

Equations (58)–(61) are the mechanical prescription for non-commuting gain/loss words. The chronological order appears in the ordered list L_μ ; changing the order changes W , and hence changes both the Born weight and the conditional Gaussian state.

The post-branch covariance is obtained from the same Wick expansion. In a Majorana basis, let γ_a be the physical Majoranas and let $\mathcal{G}_{ab}$ denote the antisymmetrized Gaussian two-point contraction. With

$$B_{\mu a} = \langle L_\mu \gamma_a \rangle, \quad (62)$$

the conditional antisymmetrized contraction is the Pfaffian Schur complement

$$\mathcal{G}'_{ab} = \mathcal{G}_{ab} + \sum_{\mu, \nu=1}^{2r} B_{\mu a} (W^{-1})_{\mu\nu} B_{\nu b}, \quad \text{Pf } W \neq 0. \quad (63)$$

Equivalently, one can evaluate the sequential rank-one updates Eqs. (30) and (31) in the actual time order. The Pfaffian formula is useful because it makes explicit that the answer is an ordered Gaussian conditioning problem, not a product of commuting scalar probabilities.

For transfer spectroscopy the conclusion is sharp. Same-type blocks may be compressed into determinant updates and a rank- r endpoint/null sector, provided the corresponding Gram matrix has rank r . A mixed non-commuting block must either be kept as an ordered exterior/interior Clifford word or embedded in a regularized ordered Majorana/Pin product. In either description, the finite singular values are read only after this ordered reduction; endpoint multiplicities are ranks of the resulting constrained subspaces, not simply the number of gain/loss events.

C. Transfer-Matrix Group: $O(2N, \mathbb{C})$ and the $U(1)$ Levi Chart

Before introducing a regulator, it is useful to state the group constraint. Let

$$\Psi = \begin{pmatrix} c \\ c^\dagger \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \quad \{\Psi_\mu, \Psi_\nu\} = \Omega_{\mu\nu}. \quad (64)$$

If an invertible Gaussian operator $\hat{V}$ acts by

$$\hat{V}\Psi\hat{V}^{-1} = \mathbb{T}[\hat{V}]\Psi, \quad (65)$$

then the canonical anticommutation relations force

$$\mathbb{T}[\hat{V}] \Omega \mathbb{T}[\hat{V}]^T = \Omega. \quad (66)$$

Thus the unprotected Nambu transfer matrix lives in the complex orthogonal group $O(2N, \mathbb{C})$, not in a free $GL(2N, \mathbb{C})$. In a Majorana basis this is the ordinary complex orthogonal constraint; the complex Nambu basis only rewrites the same bilinear form as Ω .

For a number-conserving invertible even branch the Nambu transfer is block diagonal,

$$\mathbb{T} = \begin{pmatrix} T_h & 0 \\ 0 & T_p \end{pmatrix}, \quad T_h T_p^T = \mathbb{I}, \quad T_p = T_h^{-T}. \quad (67)$$

The $U(1)$ -equivariant invertible transfer group is therefore the $GL(N, \mathbb{C})$ Levi chart parametrized by T_p . The particle/hole inverse-transpose relation is not an additional symmetry assumption; it is Eq. (66) restricted to the $U(1)$ -commuting block form.

Exact gain and loss sit outside this invertible $U(1)$ chart for a simple reason. A charge-homogeneous linear insertion is a null vector of the CAR form:

$$\hat{\chi}^\dagger = w_\chi^T \Psi, \quad w_\chi = \begin{pmatrix} 0 \\ \chi \end{pmatrix}, \quad \hat{\chi} = \bar{u}_\chi^T \Psi, \quad \bar{u}_\chi = \begin{pmatrix} \chi^* \\ 0 \end{pmatrix}, \quad w_\chi^T \Omega w_\chi = \bar{u}_\chi^T \Omega \bar{u}_\chi = 0. \quad (68)$$

Null linear Clifford elements are nilpotent and cannot be inverted. Hence no invertible regulator for a single gain or loss can remain $U(1)$ -equivariant: the regulator must mix the two charge sectors, run in $O(2N, \mathbb{C})$, and return to the charge-homogeneous operator only as a singular endpoint.

D. Moving Nambu Charts and the $O(2N, \mathbb{C})$ Regularization

There is another useful way to say the same thing. The fixed particle chart is singular because creation and annihilation are nilpotent maps. If instead one works in the full Nambu or Majorana Clifford algebra, an odd branch can be approximated by an invertible Clifford element. For a normalized mode χ , define

$$v_\varepsilon = \sqrt{\varepsilon} \hat{\chi} + \frac{1}{\sqrt{\varepsilon}} \hat{\chi}^\dagger, \quad \varepsilon > 0. \quad (69)$$

Using $\hat{\chi}^2 = (\hat{\chi}^\dagger)^2 = 0$ and $\{\hat{\chi}, \hat{\chi}^\dagger\} = 1$, one finds

$$v_\varepsilon^2 = 1, \quad v_\varepsilon^{-1} = v_\varepsilon. \quad (70)$$

Thus v_ε is not a creation operator. It is an odd Clifford, or Pin, element whose conjugation action on Nambu operators is an $O(2N, \mathbb{C})$ transformation. On the distinguished mode,

$$v_\varepsilon \hat{\chi} v_\varepsilon^{-1} = \frac{1}{\varepsilon} \hat{\chi}^\dagger, \quad v_\varepsilon \hat{\chi}^\dagger v_\varepsilon^{-1} = \varepsilon \hat{\chi}, \quad (71)$$

while modes anticommuting with χ are transformed by the corresponding orthogonal reflection, with signs fixed by the chosen Clifford ordering. Equation (71) shows exactly what the fixed particle chart calls a divergence: in the full Nambu chart it is a particle-hole reflection accompanied by a diagonal gauge $\varepsilon^{\pm 1}$.

Pure gain and loss are recovered only as projective singular limits. For gain,

$$\sqrt{\varepsilon} v_\varepsilon = \varepsilon \hat{\chi} + \hat{\chi}^\dagger \xrightarrow{\varepsilon \rightarrow 0} \hat{\chi}^\dagger, \quad (72)$$

whereas for loss one may equivalently use

$$\frac{1}{\sqrt{\varepsilon}} v_\varepsilon = \hat{\chi} + \frac{1}{\varepsilon} \hat{\chi}^\dagger \quad (73)$$

with $\varepsilon \rightarrow \infty$, or exchange $\varepsilon \leftrightarrow \varepsilon^{-1}$ in Eq. (69). More symmetrically, define

$$w_\varepsilon = \frac{1}{\sqrt{\varepsilon}} \hat{\chi} + \sqrt{\varepsilon} \hat{\chi}^\dagger, \quad w_\varepsilon^2 = 1, \quad \sqrt{\varepsilon} w_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \hat{\chi}. \quad (74)$$

The singular gain/loss maps are therefore boundary points of an invertible Pin/Bogoliubov cocycle, rather than failures of the underlying Clifford description.

This regularization is useful but not free. The physical trajectory probability is not the projective $O(2N, \mathbb{C})$ action alone. For the unrescaled operator v_ε ,

$$\begin{aligned} \text{Tr}(v_\varepsilon \rho v_\varepsilon^\dagger) &= \varepsilon \text{Tr}(\hat{\chi} \rho \hat{\chi}^\dagger) + \varepsilon^{-1} \text{Tr}(\hat{\chi}^\dagger \rho \hat{\chi}) \\ &= \varepsilon p_-(C) + \varepsilon^{-1} p_+(C), \end{aligned} \quad (75)$$

because the same-mode nilpotent cross terms contain $\hat{\chi}^2$ or $(\hat{\chi}^\dagger)^2$. After multiplying by the projective scalar $\sqrt{\varepsilon}$ appropriate to the gain limit, Eq. (75) tends to $p_+(C)$. After using the loss scaling in Eq. (74), it tends to $p_-(C)$. Consequently the moving $O(2N, \mathbb{C})$ chart regularizes the transfer direction, but the scalar cocycle must be kept separately to recover the Born weight and the trajectory free energy.

In practical terms, the two descriptions are equivalent bookkeeping choices:

$$\text{fixed particle/hole chart} \iff \text{finite block plus endpoint/null sectors}, \quad (76)$$

whereas

$$O(2N, \mathbb{C}) \text{ moving Nambu chart} \iff \begin{array}{c} \text{finite Pin/Bogoliubov cocycle plus divergent gauge} \\ \text{and scalar normalization} \end{array}. \quad (77)$$

The moving chart may be the cleaner representation for composing long trajectories with gain and loss, especially if one wants a single nonsingular Nambu transfer product. It does not eliminate the gain/loss complication; it moves it from endpoint eigenvalues into the gauge/scalar part of the cocycle.

E. Projective Measurements, Gauge Renormalization, and Particle/Hole Reciprocity

The same singular-chart issue already appears for ordinary projective occupation measurements. These are even branches, so they do not mix particle and hole sectors at finite regulator. Let $P = \chi \chi^\dagger$ be the one-body projector onto the measured particle wavefunction and define

$$D_\varepsilon(P) = \mathbb{I} - P + \varepsilon P. \quad (78)$$

The occupied and empty projectors can be approached by the invertible even branches

$$\hat{\Pi}_{1,\varepsilon} = \hat{n}_\chi + \varepsilon(1 - \hat{n}_\chi), \quad \hat{\Pi}_{0,\varepsilon} = (1 - \hat{n}_\chi) + \varepsilon \hat{n}_\chi. \quad (79)$$

For $\varepsilon > 0$, these are honest Gaussian transfer matrices and therefore compose by ordinary matrix multiplication with every other finite-regulator Gaussian factor. In the Nambu convention $\Psi = (c, c^\dagger)^T$, their block-diagonal transfer matrices are

$$\mathbb{T}_{1,\varepsilon} = \begin{pmatrix} D_\varepsilon(P)^T & 0 \\ 0 & D_\varepsilon(P)^{-1} \end{pmatrix}, \quad (80)$$

$$\mathbb{T}_{0,\varepsilon} = \begin{pmatrix} D_\varepsilon(P)^{-T} & 0 \\ 0 & D_\varepsilon(P) \end{pmatrix}. \quad (81)$$

Equivalently, on the measured mode,

$$\hat{\Pi}_{1,\varepsilon} \hat{\chi} \hat{\Pi}_{1,\varepsilon}^{-1} = \varepsilon \hat{\chi}, \quad \hat{\Pi}_{1,\varepsilon} \hat{\chi}^\dagger \hat{\Pi}_{1,\varepsilon}^{-1} = \varepsilon^{-1} \hat{\chi}^\dagger, \quad (82)$$

$$\hat{\Pi}_{0,\varepsilon} \hat{\chi} \hat{\Pi}_{0,\varepsilon}^{-1} = \varepsilon^{-1} \hat{\chi}, \quad \hat{\Pi}_{0,\varepsilon} \hat{\chi}^\dagger \hat{\Pi}_{0,\varepsilon}^{-1} = \varepsilon \hat{\chi}^\dagger. \quad (83)$$

Thus an occupied measurement produces a particle-sector pole and a hole-sector zero on the measured mode, while an empty measurement produces the opposite zero/pole pair. At every nonzero ε the particle and hole blocks are still related by CAR preservation:

$$A_h(\varepsilon)A_p(\varepsilon)^T = \mathbb{I}, \quad A_h(\varepsilon) = A_p(\varepsilon)^{-T}. \quad (84)$$

The exact projector limit destroys the full-rank chart, not the finite ε relation. On the active complement of P , the reduced particle and hole blocks remain inverse-transpose partners; on the measured direction one instead records an endpoint occupation.

The same regularized projector is equally transparent in Majorana language. Define $\gamma_{1,\chi} = \hat{\chi} + \hat{\chi}^\dagger$ and $\gamma_{2,\chi} = -i(\hat{\chi} - \hat{\chi}^\dagger)$. Then

$$\hat{n}_\chi = \frac{1 + i\gamma_{1,\chi}\gamma_{2,\chi}}{2}. \quad (85)$$

Let $B_\chi = i\gamma_{1,\chi}\gamma_{2,\chi}$, so $B_\chi^2 = 1$, and put

$$\eta_\varepsilon = \frac{1}{2} \log \frac{1}{\varepsilon}. \quad (86)$$

Then

$$\hat{\Pi}_{1,\varepsilon} = \sqrt{\varepsilon} e^{\eta_\varepsilon B_\chi}, \quad \hat{\Pi}_{0,\varepsilon} = \sqrt{\varepsilon} e^{-\eta_\varepsilon B_\chi}. \quad (87)$$

The scalar $\sqrt{\varepsilon}$ is part of the Kraus normalization and drops out of conjugation. The occupied branch acts on $(\gamma_{1,\chi}, \gamma_{2,\chi})$ by the complex orthogonal boost

$$R_{1,\varepsilon}^{(\chi)} = \begin{pmatrix} \cosh(2\eta_\varepsilon) & -i \sinh(2\eta_\varepsilon) \\ i \sinh(2\eta_\varepsilon) & \cosh(2\eta_\varepsilon) \end{pmatrix}, \quad (88)$$

and the empty branch uses $R_{0,\varepsilon}^{(\chi)} = (R_{1,\varepsilon}^{(\chi)})^{-1}$. The Nambu operators $\hat{\chi}$ and $\hat{\chi}^\dagger$ are the light-cone eigenvectors of this Majorana boost, with eigenvalues ε and ε^{-1} for the occupied outcome and the reciprocal eigenvalues for the empty outcome. Thus the Majorana and Nambu derivations are the same singular transfer written in different bases: Majorana sees a divergent boost, while Nambu diagonalizes that boost into particle/hole zero-pole factors.

This also explains what can and cannot be achieved by “renormalizing” the transfer matrix. Multiplying the Kraus operator by a scalar changes the Born weight but cancels out of the conjugation action. Multiplying the whole Nambu matrix by a scalar is not a legitimate CAR-preserving transfer matrix, except for the signs $r = \pm 1$. A legitimate number-conserving Nambu gauge has the reciprocal form

$$\mathbb{G}(R) = \begin{pmatrix} R^{-T} & 0 \\ 0 & R \end{pmatrix}, \quad \mathbb{G}(R)\Omega\mathbb{G}(R)^T = \Omega. \quad (89)$$

For example,

$$\mathbb{T}_{1,\varepsilon} = \mathbb{G}[D_\varepsilon(P)^{-1}], \quad \mathbb{T}_{0,\varepsilon} = \mathbb{G}[D_\varepsilon(P)]. \quad (90)$$

One may use such gauges to center a numerical calculation or to subtract a known regulator slope, but this does not produce a unique finite full transfer matrix in the exact limit. It only moves the zero/pole into the chart transition. The invariant data are the finite active block, the removed regulator exponents, and the endpoint occupation.

Gain and loss are the odd analogue. Their regulated Nambu matrices in Eqs. (119) and (120) are not block diagonal: the zero/pole pair appears in the particle-hole mixing blocks instead of in the diagonal projector blocks. Nevertheless the finite- ε statement is the same. The full Nambu or Majorana transfer is an ordinary $O(2N, \mathbb{C})$ matrix product. The exact $\varepsilon \rightarrow 0$ limit is a singular chart limit whose finite content is a reduced particle/hole transfer matrix plus a filled or empty endpoint sector.

The resulting interpretation of the whole monitored Gaussian circuit is: choose a finite regulator for every exact projector, gain, and loss branch; compose the associated Nambu or Majorana one-body matrices by literal ordered matrix multiplication; keep the scalar determinant/Pfaffian cocycle for the Born weight; and only then take the regulator limit. The limit returns the exact many-body Gaussian branch, but the single-particle representative is a singular-limit object rather than a globally full-rank particle-transfer matrix.

F. Majorana Transfer Form of Exact and Regularized Gain/Loss

The preceding discussion can be made completely explicit in a Majorana basis. For the mode χ , define two Majorana operators

$$\gamma_{1,\chi} = \hat{\chi} + \hat{\chi}^\dagger, \quad \gamma_{2,\chi} = -i(\hat{\chi} - \hat{\chi}^\dagger), \quad (91)$$

so that

$$\hat{\chi} = \frac{\gamma_{1,\chi} + i\gamma_{2,\chi}}{2}, \quad \hat{\chi}^\dagger = \frac{\gamma_{1,\chi} - i\gamma_{2,\chi}}{2}. \quad (92)$$

Thus exact loss and exact gain are linear Clifford insertions,

$$\hat{K}_- = \hat{\chi} = a_- \cdot \gamma, \quad \hat{K}_+ = \hat{\chi}^\dagger = a_+ \cdot \gamma, \quad (93)$$

with, in the two-dimensional subspace spanned by $(\gamma_{1,\chi}, \gamma_{2,\chi})$,

$$a_- = \frac{1}{2}(1, i), \quad a_+ = \frac{1}{2}(1, -i). \quad (94)$$

These vectors are null with respect to the complex orthogonal form:

$$a_\pm \cdot a_\pm = 0, \quad \hat{K}_\pm^2 = 0. \quad (95)$$

This is the exact Majorana-basis obstruction. Gain and loss are Majorana-linear operators, but because they are nilpotent they do not define an invertible conjugation action $\gamma_i \mapsto \hat{K}_\pm \gamma_i \hat{K}_\pm^{-1}$. Exact gain/loss is therefore a null Clifford insertion, not an element of $O(2N, \mathbb{C})$.

The regularized branch

$$v_\varepsilon = \sqrt{\varepsilon} \hat{\chi} + \frac{1}{\sqrt{\varepsilon}} \hat{\chi}^\dagger = a_\varepsilon \cdot \gamma \quad (96)$$

has

$$a_\varepsilon = \frac{1}{2} \left(\sqrt{\varepsilon} + \varepsilon^{-1/2}, i\sqrt{\varepsilon} - i\varepsilon^{-1/2} \right), \quad a_\varepsilon \cdot a_\varepsilon = 1. \quad (97)$$

Hence $v_\varepsilon^2 = 1$, as in Eq. (70), and v_ε induces an honest complex-orthogonal Majorana transfer matrix

$$v_\varepsilon \gamma_i v_\varepsilon^{-1} = \sum_j R_{ij}^{(\varepsilon)} \gamma_j, \quad R^{(\varepsilon)} \in O(2N, \mathbb{C}). \quad (98)$$

Thus the precise statement is

$$\begin{aligned} \text{exact gain/loss} &= \text{null Majorana-linear Clifford insertion,} \\ \text{regularized gain/loss} &= \text{invertible Pin element inducing } O(2N, \mathbb{C}) \text{ transfer.} \end{aligned} \quad (99)$$

The Born probabilities make clear what is, and is not, contained in this regularized transfer matrix. Let

$$n_\chi = \langle \chi | C | \chi \rangle = \frac{1 + \langle \chi | G | \chi \rangle}{2}. \quad (100)$$

For exact loss and gain,

$$p_-(C) = \text{Tr}(\hat{\chi} \rho \hat{\chi}^\dagger) = n_\chi, \quad p_+(C) = \text{Tr}(\hat{\chi}^\dagger \rho \hat{\chi}) = 1 - n_\chi. \quad (101)$$

For a same-mode regularized odd branch

$$\hat{K} = a \hat{\chi} + b \hat{\chi}^\dagger, \quad (102)$$

an even Gaussian density matrix has no parity-odd expectation values, and $\hat{\chi}^2 = (\hat{\chi}^\dagger)^2 = 0$. The cross terms therefore vanish and

$$p_K(C) = \text{Tr}(\hat{K} \rho \hat{K}^\dagger) = |a|^2 n_\chi + |b|^2 (1 - n_\chi). \quad (103)$$

For v_ε this gives

$$p_{v_\varepsilon}(C) = \varepsilon n_\chi + \varepsilon^{-1}(1 - n_\chi). \quad (104)$$

This quantity diverges in the gain limit because v_ε contains a divergent scalar gauge. The projectively rescaled gain approximation

$$\hat{K}_\varepsilon^+ = \sqrt{\varepsilon} v_\varepsilon = \varepsilon \hat{\chi} + \hat{\chi}^\dagger \quad (105)$$

has

$$p_\varepsilon^+(C) = \varepsilon^2 n_\chi + (1 - n_\chi) \xrightarrow{\varepsilon \rightarrow 0} 1 - n_\chi. \quad (106)$$

Similarly, using

$$\hat{K}_\varepsilon^- = \hat{\chi} + \varepsilon \hat{\chi}^\dagger \quad (107)$$

for loss gives

$$p_\varepsilon^-(C) = n_\chi + \varepsilon^2(1 - n_\chi) \xrightarrow{\varepsilon \rightarrow 0} n_\chi. \quad (108)$$

Therefore the $O(2N, \mathbb{C})$ transfer matrix regularizes the projective Majorana dynamics, but the physical trajectory weight lives in the scalar normalization used to take the singular gain/loss limit. Dropping that scalar would lose precisely the contribution needed for the trajectory free energy and hence for c_{eff} .

G. Regulated Products in the Majorana/Nambu Chart

The regulated Majorana chart simplifies composition at finite regulator, but only after one separates the projective transfer direction from the scalar normalization. Let

$$\gamma = (\gamma_1, \dots, \gamma_{2N})^T, \quad \{\gamma_a, \gamma_b\} = 2\delta_{ab}. \quad (109)$$

A regulated odd branch may be written as

$$\hat{K}_j(\varepsilon) = c_j(\varepsilon) \hat{u}_j(\varepsilon), \quad \hat{u}_j(\varepsilon) = u_j(\varepsilon)^T \gamma, \quad u_j(\varepsilon)^T u_j(\varepsilon) = 1. \quad (110)$$

The transpose in $u^T u$ is the complex orthogonal bilinear form, not the Hermitian norm. Since $\hat{u}_j^2 = 1$, conjugation by $\hat{u}_j$ is the complex orthogonal reflection

$$\hat{u}_j \gamma_a \hat{u}_j^{-1} = \sum_b R[u_j]_{ab} \gamma_b, \quad R[u_j]_{ab} = -\delta_{ab} + 2u_{j,a} u_{j,b}, \quad R[u_j] \in O(2N, \mathbb{C}). \quad (111)$$

Even Gaussian pieces also induce complex-orthogonal Majorana matrices. Hence a regulated trajectory with arbitrary non-commuting gains and losses becomes an ordinary ordered product of even rotations and odd reflections. The precise ordering convention matters, because conjugation is an antihomomorphism; the Nambu statement in Eqs. (128) and (129) fixes this convention explicitly. This is the real simplification: there is no separate exterior/interior algebra to commute by hand. The price is that the resulting matrix is only the projective transfer matrix.

For a single mode, the Nambu pair $\Psi_\chi = (\hat{\chi}, \hat{\chi}^\dagger)^T$ makes the regulator singularity explicit. The projectively rescaled gain and loss approximants

$$\hat{K}_\varepsilon^+ = \hat{\chi}^\dagger + \varepsilon \hat{\chi}, \quad \hat{K}_\varepsilon^- = \hat{\chi} + \varepsilon \hat{\chi}^\dagger \quad (112)$$

obey $(\hat{K}_\varepsilon^\pm)^2 = \varepsilon$, so they are invertible for $\varepsilon \neq 0$. Their conjugation actions are

$$\hat{K}_\varepsilon^+ \Psi_\chi (\hat{K}_\varepsilon^+)^{-1} = \begin{pmatrix} 0 & \varepsilon^{-1} \\ \varepsilon & 0 \end{pmatrix} \Psi_\chi, \quad (113)$$

$$\hat{K}_\varepsilon^- \Psi_\chi (\hat{K}_\varepsilon^-)^{-1} = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon^{-1} & 0 \end{pmatrix} \Psi_\chi. \quad (114)$$

Thus finite ε replaces a zero or pole in a fixed particle chart by a particle-hole exchange with singular gauge factors $\varepsilon^{\pm 1}$. These factors are not finite critical exponents.

It is useful to write the arbitrary-word formula directly in the Nambu basis. Let

$$\Psi = \begin{pmatrix} c \\ c^\dagger \end{pmatrix}, \quad \Omega = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & 0 \end{pmatrix}, \quad \{\Psi_\mu, \Psi_\nu\} = \Omega_{\mu\nu}. \quad (115)$$

For a linear form

$$\hat{\ell}(w) = w^T \Psi, \quad \hat{\ell}(w)^2 = \frac{1}{2} w^T \Omega w, \quad (116)$$

the exact gain/loss operators are precisely null vectors, $w^T \Omega w = 0$, and so have no inverse. A finite regulator replaces that null vector by a non-null vector. If $w^T \Omega w \neq 0$, conjugation by the normalized Pin element induces

$$\hat{\ell}(w) \Psi \hat{\ell}(w)^{-1} = \mathbb{R}[w] \Psi, \quad \mathbb{R}[w] = -\mathbb{I} + \frac{2\Omega w w^T}{w^T \Omega w}, \quad \mathbb{R}[w] \Omega \mathbb{R}[w]^T = \Omega. \quad (117)$$

The factor Ωw , rather than w , appears because w labels a linear functional $w^T \Psi$. Equation (117) is the Nambu version of the Majorana reflection in Eq. (111).

For a mode χ , with $\hat{\chi}^\dagger = \sum_i \chi_i c_i^\dagger$ and $\hat{\chi} = \sum_i \chi_i^* c_i$, define

$$w_+(\varepsilon) = \begin{pmatrix} \varepsilon^{1/2} \chi^* \\ \varepsilon^{-1/2} \chi \end{pmatrix}, \quad w_-(\varepsilon) = \begin{pmatrix} \varepsilon^{-1/2} \chi^* \\ \varepsilon^{1/2} \chi \end{pmatrix}. \quad (118)$$

Then $\hat{\ell}[w_+(\varepsilon)]$ tends projectively to gain as $\varepsilon \rightarrow 0$, while $\hat{\ell}[w_-(\varepsilon)]$ tends projectively to loss. Since $w_\pm^T \Omega w_\pm = 2$, the Nambu reflection matrices are

$$\mathbb{R}_+(\chi, \varepsilon) = \begin{pmatrix} -(\mathbb{I} - P) & \varepsilon^{-1} \chi \chi^T \\ \varepsilon \chi^* \chi^\dagger & -(\mathbb{I} - P^T) \end{pmatrix}, \quad (119)$$

$$\mathbb{R}_-(\chi, \varepsilon) = \begin{pmatrix} -(\mathbb{I} - P) & \varepsilon \chi \chi^T \\ \varepsilon^{-1} \chi^* \chi^\dagger & -(\mathbb{I} - P^T) \end{pmatrix}, \quad (120)$$

where $P = \chi \chi^\dagger$. For $N = 1$ these reduce to Eqs. (113) and (114). The off-diagonal blocks are the charge-breaking regulator pieces. The full matrix remains exactly CAR-orthogonal; if one drops either the divergent or vanishing off-diagonal block before taking the limit, the missing product $\varepsilon^{-1} \varepsilon P$ is precisely the occupation constraint that has to be restored by hand in the fixed particle/hole chart.

This statement becomes concrete when the neighboring even segment is number-conserving. Write

$$\mathbb{A} = \begin{pmatrix} A_h & 0 \\ 0 & A_p \end{pmatrix}, \quad A_h A_p^T = \mathbb{I}, \quad (121)$$

and set $P_h = P = \chi \chi^\dagger$, $P_p = P^T$. For an odd insertion on the right side of this even transfer, ordinary Nambu multiplication gives

$$\mathbb{A} \mathbb{R}_+(\chi, \varepsilon) = \begin{pmatrix} -A_h(\mathbb{I} - P_h) & \varepsilon^{-1} A_h \chi \chi^T \\ \varepsilon A_p \chi^* \chi^\dagger & -A_p(\mathbb{I} - P_p) \end{pmatrix}, \quad (122)$$

$$\mathbb{A} \mathbb{R}_-(\chi, \varepsilon) = \begin{pmatrix} -A_h(\mathbb{I} - P_h) & \varepsilon A_h \chi \chi^T \\ \varepsilon^{-1} A_p \chi^* \chi^\dagger & -A_p(\mathbb{I} - P_p) \end{pmatrix}. \quad (123)$$

Thus the diagonal particle/hole blocks have a finite, rank-deficient limit, while one charge-mixing block vanishes and the other diverges. For a right insertion the fixed-chart finite blocks are

$$A_h \mapsto A_h(\mathbb{I} - P_h), \quad A_p \mapsto A_p(\mathbb{I} - P_p), \quad (124)$$

up to the unimportant reflection sign. For a left insertion the same calculation gives left multiplication by the complementary projectors,

$$A_h \mapsto (\mathbb{I} - P_h) A_h, \quad A_p \mapsto (\mathbb{I} - P_p) A_p. \quad (125)$$

Inside a longer word the mode appearing in P_h or P_p is transported by the adjacent even matrices, exactly as in Eqs. (39) and (40). The exact gain/loss operation is therefore recovered from the regulated product as a reduced finite particle/hole transfer plus endpoint occupation data: gain and loss have the same finite rank drop in the diagonal blocks, but gain has the ε^{-1} pole in the particle-hole block and records a filled endpoint, while loss has the opposite pole and records an empty endpoint. A fixed particle chart or fixed hole chart sees this as a zero or infinity; the full Nambu chart sees it as a singular gauge direction of an otherwise ordinary $O(2N, \mathbb{C})$ matrix product.

Now let the exact trajectory word be an arbitrary Gaussian word made of even Gaussian pieces and exact gain/loss insertions. Its regulated version can be written as

$$\hat{K}_{\xi, \varepsilon} = s_{\xi}(\varepsilon) \hat{E}_m \hat{\ell}[w_m(\varepsilon)] \hat{E}_{m-1} \cdots \hat{\ell}[w_1(\varepsilon)] \hat{E}_0, \quad (126)$$

where each $\hat{E}_j$ is an invertible even Gaussian operator, not assumed to be number conserving, and $s_{\xi}(\varepsilon)$ is the scalar projective normalization. If

$$\hat{E}_j \Psi \hat{E}_j^{-1} = \mathbb{A}_j \Psi, \quad \mathbb{A}_j \Omega \mathbb{A}_j^T = \Omega, \quad (127)$$

then, with the convention $\hat{V} \Psi \hat{V}^{-1} = \mathbb{T}[\hat{V}] \Psi$, the map $\hat{V} \mapsto \mathbb{T}[\hat{V}]$ is an antihomomorphism:

$$\mathbb{T}[\hat{V}_2 \hat{V}_1] = \mathbb{T}[\hat{V}_1] \mathbb{T}[\hat{V}_2]. \quad (128)$$

Therefore the regulated word has the transfer matrix

$$\mathbb{T}_{\xi, \varepsilon} = \mathbb{A}_0 \mathbb{R}[w_1(\varepsilon)] \mathbb{A}_1 \cdots \mathbb{R}[w_m(\varepsilon)] \mathbb{A}_m, \quad \mathbb{T}_{\xi, \varepsilon} \Omega \mathbb{T}_{\xi, \varepsilon}^T = \Omega. \quad (129)$$

This is the clean Nambu statement for non-commuting gain/loss strings: all ordering is ordinary matrix ordering in Eq. (129). No separate rule allows the gain factors to be commuted past the loss factors.

H. Laurent Limit of an Arbitrary Regulated Word

The preceding product formula is exact but it is not yet the exact gain/loss endpoint. For every $\varepsilon > 0$ the regulated Nambu, or Majorana, representative is an honest $O(2N, \mathbb{C})$ matrix and products are literal ordered matrix products. The singular endpoint is recovered only after passing to a graph/covariance chart and separating the scalar normalization. Taking the entrywise $\varepsilon \rightarrow 0$ limit of a raw particle or hole block is generally the wrong operation.

A useful way to organize an arbitrary exact word is by its running charge. Read the odd insertions in chronological order and define

$$Q_j = \#\{\text{gains among } 1, \dots, j\} - \#\{\text{losses among } 1, \dots, j\}, \quad Q_0 = 0. \quad (130)$$

Cut the word whenever Q_j returns to its running minimum. These cuts isolate neutral blocks that can close internally from charge-changing tails. For a neutral block B with nonzero contraction amplitude, the exact operator has the form

$$\hat{K}_B = \lambda_B \hat{K}_{T_B}, \quad \lambda_B \neq 0, \quad (131)$$

where $\hat{K}_{T_B}$ is an even Gaussian operator with a finite one-body transfer T_B on the active graph chart. For a block with net charge $q > 0$,

$$\hat{K}_B = \lambda_B \hat{c}^\dagger(\xi_q) \cdots \hat{c}^\dagger(\xi_1) \hat{K}_{T_B}, \quad (132)$$

and for $q < 0$ the exterior product is replaced by an ordered interior product of $|q|$ annihilation modes. Thus an unbalanced block is not a matrix alone; it is endpoint orbitals plus one matrix plus a scalar.

For a balanced mixed block, the scalar and the finite graph transfer are fixed by the causal loss-gain Gram. Let i label gains with incoming wavefunctions α_i , let j label losses with outgoing wavefunctions β_j , and let M_{ji} be the even one-body propagation from gain i to loss j through the pieces lying between them. The internal contraction matrix is

$$\tilde{G}_{ji} = \Theta(t_j^\ell > t_i^g) \langle \beta_j | M_{ji} | \alpha_i \rangle. \quad (133)$$

The causal step function is the algebraic memory of the operator order: same-type pairings vanish by isotropy of the gain and loss null subspaces, while a loss can only contract a gain that occurs earlier in the word. If $\det \tilde{G} \neq 0$, then

$$\lambda_B = \pm \det \tilde{G}, \quad (134)$$

where the sign is the fermionic ordering sign. This scalar is part of the Born-weight cocycle. It is not determined by the projective one-body transfer matrix.

After partitioning the finite- ε Nambu graph into external propagating legs b , future-gain legs fg , and loss legs l , the Schur complement gives the dressed master formula

$$\frac{T_B}{\lambda_B} = M_{bb} - W_{fg} \tilde{G}^{-1} W_l^\dagger. \quad (135)$$

Equivalently, if the regulated one-body graph is written in block form with the lower-right graph block $A_{dd}(\varepsilon)$, the finite object that converges is

$$\lim_{\varepsilon \rightarrow 0} (A_{dd}(\varepsilon)^T)^{-1} = M_{bb} - W_{fg} \tilde{G}^{-1} W_l^\dagger. \quad (136)$$

This is the same dressed-projector formula obtained by direct Choi/covariance contraction. The raw particle block $A_{cc}(\varepsilon)$ can diverge, and its projectively rescaled limit is typically only a flag. The block in Eq. (136), not the raw particle block, is the finite transfer chart relevant for active-sector singular data.

The same-mode identities are a useful check:

$$\hat{\chi} \hat{\chi}^\dagger = \mathbb{I} - \hat{n}_\chi, \quad \hat{\chi}^\dagger \hat{\chi} = \hat{n}_\chi. \quad (137)$$

They are the one-mode version of Eq. (135). Same-type gain or loss blocks instead grow endpoint occupation constraints. For gains the Born factor is the empty-space Gram $\det[X^\dagger(\mathbb{I} - C)X]$; for losses it is the occupied-space Gram $\det[Y^\dagger CY]$. Non-orthogonal wavefunctions are handled by these Gram determinants and the corresponding Schur complements; small or zero determinants signal redundant, blocked, or ill-conditioned exact branches, not additional finite Lyapunov gaps.

The finite spectrum is extracted from singular-value asymptotics only after this separation. If $s_a(\varepsilon)$ are singular values of the regulated product, define

$$\log s_a(\varepsilon) = \kappa_a \log \frac{1}{\varepsilon} + \rho_a + o(1). \quad (138)$$

CAR preservation pairs nonzero slopes κ_a with opposite slopes. Those directions are endpoint or gauge sectors. The regulator-stable offsets ρ_a with $\kappa_a = 0$, together with the scalar λ_B and the filled/empty endpoint orbitals, are the complete singular data. This is why the regulated Majorana basis simplifies composition but does not remove endpoint or Born-weight bookkeeping.

I. Case-by-Case Regulated Endpoint Limits

Here is the same regulator limit written mechanically for the elementary cases. In this subsection the regulator is denoted by δ , to avoid confusing it with the exterior multiplication operator ε_x . Work first in a chart where all even pieces have already transported the odd wavefunctions to the relevant edge. The even pieces are restored by replacing the displayed modes by the transported modes of Eqs. (44) and (51), and by multiplying the active block by the total even one-body transfer. Define

$$\hat{k}_\delta^+(x) = \hat{c}^\dagger(x) + \delta \hat{c}(x), \quad \hat{k}_\delta^-(y) = \hat{c}(y) + \delta \hat{c}^\dagger(y). \quad (139)$$

For normalized x and y , $(\hat{k}_\delta^\pm)^2 = \delta$, so $\delta^{-1/2} \hat{k}_\delta^\pm$ is an invertible Pin element at every $\delta \neq 0$. Thus the finite- δ one-body Nambu/Majorana representative composes by ordinary matrix multiplication. The exact branch is the leading coefficient of the operator expansion as $\delta \rightarrow 0$.

1. *One gain.* The regulated operator is

$$\hat{k}_\delta^+(x) = \hat{c}^\dagger(x) + \delta \hat{c}(x) = \hat{c}^\dagger(x) + O(\delta). \quad (140)$$

Thus the exact limit is the exterior map $\varepsilon_x : \Psi_q \mapsto x \wedge \Psi_q$. It is not a scalar and not a same-charge transfer. For an incoming gauge-invariant Gaussian covariance C ,

$$p_{+,x}(C) = \text{Tr}[\hat{c}^\dagger(x) \rho \hat{c}(x)] = x^\dagger (\mathbb{I} - C)x. \quad (141)$$

If $x = \|x\| \hat{x}$, the exact operator is $\|x\| \hat{c}^\dagger(\hat{x})$. The scalar $\|x\|$ belongs to the branch amplitude, while the active particle chart removes the filled endpoint,

$$T_{\text{act}}^{+,x} = T_{\text{even}}(\mathbb{I} - P_{\hat{x}}), \quad P_{\hat{x}} = \hat{x} \hat{x}^\dagger. \quad (142)$$

The finite- δ $O(2N, \mathbb{C})$ matrix has one δ^{-1} direction and one reciprocal δ direction in the particle/hole pair of x . Those directions are the filled endpoint sector, not finite transfer gaps.

2. *Two gains.* For two gains in chronological order x_1 then x_2 ,

$$\begin{aligned}\hat{k}_\delta^+(x_2)\hat{k}_\delta^+(x_1) &= \hat{c}^\dagger(x_2)\hat{c}^\dagger(x_1) + \delta[\hat{c}^\dagger(x_2)\hat{c}(x_1) + \hat{c}(x_2)\hat{c}^\dagger(x_1)] + \delta^2\hat{c}(x_2)\hat{c}(x_1) \\ &= \hat{c}^\dagger(x_2)\hat{c}^\dagger(x_1) + O(\delta).\end{aligned}\quad (143)$$

The middle term contains the contraction $\hat{c}(x_2)\hat{c}^\dagger(x_1) = \langle x_2|x_1 \rangle - \hat{c}^\dagger(x_1)\hat{c}(x_2)$, but it is still suppressed by δ . Therefore, when $x_1 \wedge x_2 \neq 0$, the exact endpoint is a two-dimensional filled subspace. The Born factor is

$$p_{++,X}(C) = \det \begin{pmatrix} x_1^\dagger(\mathbb{I} - C)x_1 & x_1^\dagger(\mathbb{I} - C)x_2 \\ x_2^\dagger(\mathbb{I} - C)x_1 & x_2^\dagger(\mathbb{I} - C)x_2 \end{pmatrix}, \quad X = (x_1, x_2). \quad (144)$$

If $X = UR$ is a QR decomposition of the independent endpoint span, then

$$\hat{c}^\dagger(x_2)\hat{c}^\dagger(x_1) = \det(R) \hat{c}^\dagger(u_2)\hat{c}^\dagger(u_1), \quad P_X = UU^\dagger. \quad (145)$$

The active transfer is $T_{\text{even}}(\mathbb{I} - P_X)$, and the Nambu singular spectrum contains two reciprocal endpoint pairs. If $\text{rank } X < 2$, the exact two-gain branch vanishes by antisymmetry; the regulator then selects a δ -suppressed different word and the exact branch should be treated as blocked or censored.

3. *N gains.* For N gains, put $X = (x_1, \dots, x_N)$ and write

$$\hat{K}_{+,N}(\delta) = \hat{k}_\delta^+(x_N) \cdots \hat{k}_\delta^+(x_1) = \sum_{S \subset \{1, \dots, N\}} \delta^{|S|} \prod_{j=N}^1 \hat{d}_j(S), \quad (146)$$

where

$$\hat{d}_j(S) = \begin{cases} \hat{c}(x_j), & j \in S, \\ \hat{c}^\dagger(x_j), & j \notin S. \end{cases} \quad (147)$$

The leading exact word is therefore

$$\hat{K}_{+,N}(0) = \hat{c}^\dagger(x_N) \cdots \hat{c}^\dagger(x_1). \quad (148)$$

If X has rank $r = N$ and $X = UR$, then

$$\hat{K}_{+,N}(0) = \det(R) \hat{c}^\dagger(u_N) \cdots \hat{c}^\dagger(u_1), \quad P_X = UU^\dagger. \quad (149)$$

The branch fills the endpoint subspace $\text{im } P_X$, the Born factor is

$$p_{+,X}(C) = \det[X^\dagger(\mathbb{I} - C)X], \quad (150)$$

and the active particle transfer is

$$T_{\text{act}}^{+X} = T_{\text{even}}(\mathbb{I} - P_X). \quad (151)$$

The finite-regulator one-body product has N reciprocal zero/pole pairs after the finite change of endpoint basis $X \mapsto U$. Rank deficiency of X , or singularity of the empty Gram $X^\dagger(\mathbb{I} - C)X$, means that the leading exact branch has zero wedge amplitude or zero Born probability.

4. *One loss.* The loss branch is the particle-hole dual:

$$\hat{k}_\delta^-(y) = \hat{c}(y) + \delta\hat{c}^\dagger(y) = \hat{c}(y) + O(\delta). \quad (152)$$

The exact map is the interior product ι_y . Its Born factor is

$$p_{-,y}(C) = \text{Tr}[\hat{c}(y)\rho\hat{c}^\dagger(y)] = y^\dagger C y. \quad (153)$$

If $y = \|y\|\hat{y}$, the active finite chart is most naturally the hole chart:

$$T_{\text{act},h}^{-y} = T_{\text{even},h}(\mathbb{I} - P_{\hat{y}}^T). \quad (154)$$

The particle chart sees the reciprocal endpoint pole. The endpoint occupation label is empty rather than filled.

5. *Two losses.* For two losses,

$$\begin{aligned}\hat{k}_\delta^-(y_2)\hat{k}_\delta^-(y_1) &= \hat{c}(y_2)\hat{c}(y_1) + \delta[\hat{c}(y_2)\hat{c}^\dagger(y_1) + \hat{c}^\dagger(y_2)\hat{c}(y_1)] + \delta^2\hat{c}^\dagger(y_2)\hat{c}^\dagger(y_1) \\ &= \hat{c}(y_2)\hat{c}(y_1) + O(\delta).\end{aligned}\quad (155)$$

The exact endpoint is a two-dimensional empty subspace if $y_1 \wedge y_2 \neq 0$. With $Y = (y_1, y_2)$,

$$p_{--,Y}(C) = \det \begin{pmatrix} y_1^\dagger C y_1 & y_1^\dagger C y_2 \\ y_2^\dagger C y_1 & y_2^\dagger C y_2 \end{pmatrix}. \quad (156)$$

After $Y = VR$, the endpoint projector is $P_Y = VV^\dagger$, the active hole transfer is $T_{\text{even},h}(\mathbb{I} - P_Y^T)$, and there are two reciprocal regulator pairs. Rank deficiency or a singular occupied Gram means that the exact two-loss branch is redundant or blocked.

6. *M losses.* For M losses,

$$\hat{K}_{-,M}(\delta) = \hat{k}_\delta^-(y_M) \cdots \hat{k}_\delta^-(y_1) = \sum_{S \subset \{1, \dots, M\}} \delta^{|S|} \prod_{j=M}^1 \hat{e}_j(S), \quad (157)$$

with

$$\hat{e}_j(S) = \begin{cases} \hat{c}^\dagger(y_j), & j \in S, \\ \hat{c}(y_j), & j \notin S. \end{cases} \quad (158)$$

Thus

$$\hat{K}_{-,M}(0) = \hat{c}(y_M) \cdots \hat{c}(y_1). \quad (159)$$

For independent $Y = VR$, this is the oriented interior product on $\text{im } P_Y$, multiplied by the determinant of the change of basis. The Born factor and active hole transfer are

$$p_{-,Y}(C) = \det[Y^\dagger CY], \quad T_{\text{act},h}^{-Y} = T_{\text{even},h}(\mathbb{I} - P_Y^T). \quad (160)$$

The regulator produces M reciprocal endpoint pairs, now labelled as empty sectors.

7. *One gain and one loss.* There are two inequivalent orders. If the gain occurs first and the loss later, the operator product is

$$\begin{aligned}\hat{k}_\delta^-(\beta)\hat{k}_\delta^+(\alpha) &= \hat{c}(\beta)\hat{c}^\dagger(\alpha) + \delta[\hat{c}(\beta)\hat{c}(\alpha) + \hat{c}^\dagger(\beta)\hat{c}^\dagger(\alpha)] + \delta^2\hat{c}^\dagger(\beta)\hat{c}(\alpha) \\ &= \langle \beta | \alpha \rangle - \hat{c}^\dagger(\alpha)\hat{c}(\beta) + O(\delta).\end{aligned}\quad (161)$$

With an even segment M between the gain and the loss, replace α by $M\alpha$ in the contraction; the causal Gram is the scalar $\tilde{G} = \langle \beta | M | \alpha \rangle$. If $\tilde{G} \neq 0$, this is the one-pair version of Eq. (135): a closed neutral block with scalar $\lambda = \tilde{G}$ and a dressed rank-one subtraction in the active graph chart. For $\beta = \alpha$ and $M = \mathbb{I}$ it reduces to $\mathbb{I} - \hat{n}_\alpha$, the empty projector. The Born factor for the exact two-step branch is

$$p_{-+}^{(0)} = [\alpha^\dagger(\mathbb{I} - C)\alpha][\beta^\dagger C\beta] + |\beta^\dagger(\mathbb{I} - C)\alpha|^2. \quad (162)$$

If the loss occurs first and the gain later, then

$$\begin{aligned}\hat{k}_\delta^+(\alpha)\hat{k}_\delta^-(\beta) &= \hat{c}^\dagger(\alpha)\hat{c}(\beta) + \delta[\hat{c}^\dagger(\alpha)\hat{c}^\dagger(\beta) + \hat{c}(\alpha)\hat{c}(\beta)] + \delta^2\hat{c}(\alpha)\hat{c}^\dagger(\beta) \\ &= \hat{c}^\dagger(\alpha)\hat{c}(\beta) + O(\delta).\end{aligned}\quad (163)$$

This is neutral in charge but not a scalar multiple of an invertible even Gaussian transfer. It is an exterior–interior map; for the same normalized mode it is $\hat{n}_\alpha$, the occupied projector. Its Born factor is

$$p_{+-}^{(0)} = [\beta^\dagger C\beta][\alpha^\dagger(\mathbb{I} - C)\alpha] + |\alpha^\dagger C\beta|^2. \quad (164)$$

The two orders coincide only in special cases. The regulator has not made gain and loss commute; it has made their finite- δ composition an ordinary matrix product whose endpoint remembers the order.

8. *N gains and M losses.* For a general chronological word, let $\sigma_j = +$ denote a gain and $\sigma_j = -$ denote a loss. Define

$$d_j = \begin{cases} \hat{c}^\dagger(z_j), & \sigma_j = +, \\ \hat{c}(z_j), & \sigma_j = -, \end{cases} \quad \bar{d}_j = \begin{cases} \hat{c}(z_j), & \sigma_j = +, \\ \hat{c}^\dagger(z_j), & \sigma_j = -. \end{cases} \quad (165)$$

The regulated word is exactly

$$\hat{K}_\sigma(\delta) = \prod_{j=r}^1 (d_j + \delta \bar{d}_j) = \sum_{S \subset \{1, \dots, r\}} \delta^{|S|} \prod_{j=r}^1 d_j(S), \quad d_j(S) = \begin{cases} \bar{d}_j, & j \in S, \\ d_j, & j \notin S. \end{cases} \quad (166)$$

Therefore

$$\hat{K}_\sigma(\delta) = \hat{K}_\sigma(0) + O(\delta), \quad \hat{K}_\sigma(0) = d_r \cdots d_1. \quad (167)$$

For a $U(1)$ -invariant incoming state, a single flipped branch changes the net charge by two, so the Born probability satisfies

$$\text{Tr}[\hat{K}_\sigma(\delta) \rho \hat{K}_\sigma(\delta)^\dagger] = \text{Tr}[\hat{K}_\sigma(0) \rho \hat{K}_\sigma(0)^\dagger] + O(\delta^2), \quad (168)$$

unless the leading exact probability vanishes, in which case the limit is nonuniform and the exact branch is blocked.

The leading word $d_r \cdots d_1$ is then reduced by the Clifford relation $\iota_u \varepsilon_v + \varepsilon_v \iota_u = \langle u|v \rangle$. Equivalently, build the ordered Pfaffian matrix of Eq. (61). For balanced causal blocks with $N = M$ and nonsingular causal Gram $\tilde{G}$,

$$\hat{K}_\sigma(0) = \lambda \hat{K}_T, \quad \lambda = \pm \det \tilde{G}, \quad \frac{T}{\lambda} = M_{bb} - W_{fg} \tilde{G}^{-1} W_l^\dagger. \quad (169)$$

More generally, let r_c be the rank of the causal contraction problem after the chronological word has been cut into the blocks specified by Eq. (130). The reduced word then has $N - r_c$ filled endpoint orbitals and $M - r_c$ empty endpoint orbitals/interior maps. Thus even when $N = M$, an anticausal or rank-deficient ordering need not reduce to a matrix alone. In all cases the finite- δ one-body matrices compose literally; the limit is not the raw matrix product but the triple

$$(\text{scalar Pfaffian/determinant}, \quad \text{finite graph transfer}, \quad \text{filled/empty endpoint sectors}). \quad (170)$$

The Born weight of the same regulated product is not determined by the projective one-body matrix $\mathbb{T}_{\xi, \varepsilon}$, or by its Majorana-basis representative, alone. For one branch and an even Gaussian density matrix, define

$$M_{ab} = \text{Tr}(\rho \gamma_a \gamma_b). \quad (171)$$

Equation (110) gives

$$p_j(\varepsilon) = \text{Tr}[\hat{K}_j(\varepsilon) \rho \hat{K}_j(\varepsilon)^\dagger] = |c_j(\varepsilon)|^2 u_j(\varepsilon)^\dagger M u_j(\varepsilon). \quad (172)$$

For a same-mode number-conserving state and $\hat{K} = a\hat{\chi} + b\hat{\chi}^\dagger$, this reduces to

$$p_{a,b}(C) = |a|^2 n_\chi + |b|^2 (1 - n_\chi), \quad n_\chi = \langle \chi | C | \chi \rangle. \quad (173)$$

Therefore

$$p_\varepsilon^+(C) = 1 - n_\chi + \varepsilon^2 n_\chi, \quad p_\varepsilon^-(C) = n_\chi + \varepsilon^2 (1 - n_\chi), \quad (174)$$

which tends to the exact gain/loss probabilities in Eq. (101).

For an ordered string of regulated odd insertions,

$$\hat{K}_{\xi, \varepsilon} = \left[\prod_{j=1}^r c_j(\varepsilon) \right] \hat{u}_r(\varepsilon) \cdots \hat{u}_1(\varepsilon). \quad (175)$$

Let

$$L_1, \dots, L_{2r} = \hat{u}_1^\dagger, \dots, \hat{u}_r^\dagger, \hat{u}_r, \dots, \hat{u}_1 \quad (176)$$

be the ordered list entering $\hat{K}_{\xi,\varepsilon}^\dagger \hat{K}_{\xi,\varepsilon}$. With

$$W_{\mu\nu}^{(\varepsilon)} = \begin{cases} \langle L_\mu L_\nu \rangle, & \mu < \nu, \\ -\langle L_\nu L_\mu \rangle, & \mu > \nu, \\ 0, & \mu = \nu, \end{cases} \quad (177)$$

Wick's theorem gives

$$p_{\xi,\varepsilon} = \text{Tr}(\hat{K}_{\xi,\varepsilon} \rho \hat{K}_{\xi,\varepsilon}^\dagger) = \left| \prod_{j=1}^r c_j(\varepsilon) \right|^2 \text{Pf } W^{(\varepsilon)}. \quad (178)$$

Even Gaussian segments are included by inserting their own Bogoliubov transfer matrices into Eq. (129) and by keeping their determinant/scalar normalizations in the same scalar cocycle. Thus the projective matrix product and the scalar/Pfaffian normalization are two pieces of one trajectory amplitude, not interchangeable quantities.

For fixed finite string length, the exact word is recovered as

$$\hat{K}_{\xi,\varepsilon} = \hat{K}_{\xi,0} + O(\varepsilon), \quad (179)$$

where $\hat{K}_{\xi,0}$ is the chronological word of exact $\hat{\chi}^\dagger$'s and $\hat{\chi}$'s. If the incoming Gaussian state is $U(1)$ -invariant, terms with a single flipped branch carry the wrong charge and do not interfere with the exact word, so generically

$$p_{\xi,\varepsilon} = p_{\xi,0} + O(\varepsilon^2). \quad (180)$$

Without $U(1)$ symmetry the convergence is still to the exact word, but the leading correction can be $O(\varepsilon)$. If $p_{\xi,0} = 0$, the limit is nonuniform: after normalization the regulated state is controlled by the first nonzero ε -suppressed word, not by the blocked exact branch. Such events should be treated as censored exact gain/loss branches.

Finally, let $s_a(\varepsilon)$ be singular values of the projective product $\mathbb{T}_{\xi,\varepsilon}$. Regulator directions are detected by

$$\kappa_a = \lim_{\varepsilon \rightarrow 0} \frac{\log s_a(\varepsilon)}{\log(1/\varepsilon)}. \quad (181)$$

Nonzero κ_a are endpoint/gauge sectors. The finite transfer gaps are the regulator-stable singular data, or Fock-space combinations whose $\log \varepsilon$ dependence cancels after the scalar normalization in Eq. (178) is included. This is why a regulated Majorana basis simplifies composition but does not remove endpoint bookkeeping or Born-weight bookkeeping.

J. Dressed Projectors from the Single-Copy Nambu Regulator

The dressed projector in the Choi-contraction calculation can be obtained without passing to the doubled superoperator representation. The only essential point is the order of limits: first compose the finite-regulator Nambu word, or equivalently transport the regulated odd linear form through the later even factor; only then take the singular endpoint limit.

In this subsection it is cleaner to remove the antihomomorphism ambiguity by using the circuit-order matrix

$$L[\hat{V}] \equiv \mathbb{T}[\hat{V}]^T, \quad L[\hat{V}_2 \hat{V}_1] = L[\hat{V}_2] L[\hat{V}_1]. \quad (182)$$

This is only a local convention; elsewhere the note keeps the $\mathbb{T}$ -convention of Eq. (128). For a gain in mode χ , define the null Nambu covectors

$$w_\chi = \begin{pmatrix} 0 \\ \chi \end{pmatrix}, \quad \bar{u}_\chi = \begin{pmatrix} \chi^* \\ 0 \end{pmatrix}, \quad w_\chi^T \Omega \bar{u}_\chi = 1, \quad (183)$$

and the normalized non-null regulator

$$\zeta_\varepsilon = \varepsilon^{-1/2} w_\chi + \varepsilon^{1/2} \bar{u}_\chi, \quad \zeta_\varepsilon^T \Omega \zeta_\varepsilon = 2. \quad (184)$$

The corresponding Pin element is $\hat{g}(\varepsilon) = \hat{\ell}(\zeta_\varepsilon)$, and its reflection is

$$L_g(\varepsilon) = \zeta_\varepsilon^T \Omega - \mathbb{I}, \quad L_g(\varepsilon) \Omega L_g(\varepsilon)^T = \Omega. \quad (185)$$

The projectively normalized operator used below, $\hat{k}_\varepsilon^+(\chi) = \hat{c}^\dagger(\chi) + \varepsilon \hat{c}(\chi)$, is $\sqrt{\varepsilon} \hat{\ell}(\zeta_\varepsilon)$. The scalar $\sqrt{\varepsilon}$ belongs to the amplitude cocycle and is not visible in the projective matrix L_g .

Now sandwich the regulated gain by two invertible even Gaussian pieces, $\hat{K}(\varepsilon) = \hat{V}_2 \hat{g}(\varepsilon) \hat{V}_1$, and let $L_i = L[\hat{V}_i]$. Orthogonality gives

$$L_K(\varepsilon) = L_2 L_g(\varepsilon) L_1 = a(\varepsilon) b(\varepsilon)^T \Omega - L_0, \quad a = L_2 \zeta_\varepsilon, \quad b = L_1^{-1} \zeta_\varepsilon, \quad L_0 = L_2 L_1. \quad (186)$$

This is an exact finite- ε identity in $O(2N, \mathbb{C})$. Its projectively rescaled matrix limit is

$$\varepsilon L_K(\varepsilon) \xrightarrow{\varepsilon \rightarrow 0} (L_2 w_\chi)(L_1^{-1} w_\chi)^T \Omega, \quad (187)$$

which has rank one. That rank-one flag is not a valid transfer matrix on the full graph. The finite dressed projector is obtained only after moving to the appropriate graph/covariance chart, or equivalently after transporting the regulated null vector before taking the endpoint limit.

Let $\hat{V}$ and $\hat{W}$ be number-conserving even Gaussian operators with particle transfer matrices T and S , using the convention

$$T_p(\hat{W} \hat{V}) = T_p(\hat{V}) T_p(\hat{W}) = TS.$$

For a one-particle wavefunction x , write

$$\hat{c}^\dagger(x) = \sum_i x_i \hat{c}_i^\dagger, \quad \hat{c}(x) = \sum_i x_i^* \hat{c}_i. \quad (188)$$

The even operator $\hat{W}$ acts on creation and annihilation covectors as

$$\hat{W} \hat{c}^\dagger(\chi) \hat{W}^{-1} = \hat{c}^\dagger(S^\dagger \chi), \quad \hat{W} \hat{c}(\chi) \hat{W}^{-1} = \hat{c}(S^{-1} \chi). \quad (189)$$

Equivalently, with $\Psi = (c, c^\dagger)^T$ and $\hat{\ell}(w) = w^T \Psi$, the gain regulator has Nambu covector

$$w_\varepsilon^+(\chi) = \begin{pmatrix} \varepsilon \chi^* \\ \chi \end{pmatrix}, \quad \hat{\ell}[w_\varepsilon^+(\chi)] = \hat{c}^\dagger(\chi) + \varepsilon \hat{c}(\chi). \quad (190)$$

Finite Nambu matrix composition transports this covector to

$$\tilde{w}_\varepsilon^+(\chi; S) = \begin{pmatrix} \varepsilon(S^{-1} \chi)^* \\ S^\dagger \chi \end{pmatrix}. \quad (191)$$

Thus the finite Nambu regulator for a gain insertion,

$$\hat{k}_\varepsilon^+(\chi) = \hat{c}^\dagger(\chi) + \varepsilon \hat{c}(\chi), \quad [\hat{k}_\varepsilon^+(\chi)]^2 = \varepsilon, \quad (192)$$

gives the exact finite-regulator identity

$$\hat{W} \hat{k}_\varepsilon^+(\chi) \hat{V} = [\hat{c}^\dagger(x) + \varepsilon \hat{c}(y)] \hat{W} \hat{V}, \quad (193)$$

$$x = S^\dagger \chi, \quad y = S^{-1} \chi, \quad y^\dagger x = 1. \quad (194)$$

Equation (193) is the single-copy Nambu version of the dressing rule. After a nonunitary S , the regulator partner is not the Hermitian conjugate wavefunction of the dominant creation mode. The pair (x, y) is instead biorthogonal. This is why freezing the undressed projector P_χ before composing with S gives the wrong finite chart.

Since the projective limit is controlled by the creation component x , put

$$\hat{x} = \frac{x}{\|x\|}, \quad P_{\hat{x}} = \hat{x} \hat{x}^\dagger = \frac{S^\dagger \chi \chi^\dagger S}{\chi^\dagger S S^\dagger \chi}. \quad (195)$$

The scalar $\|x\|$ belongs to the branch normalization. It affects the Born weight but not the finite projective transfer block. The exact gain branch is

$$\hat{W} \hat{\chi}^\dagger \hat{V} = \|x\| \hat{c}^\dagger(\hat{x}) \hat{W} \hat{V}. \quad (196)$$

Using $\hat{c}^\dagger(\hat{x}) = \hat{c}^\dagger(\hat{x})(\mathbb{I} - \hat{n}_{\hat{x}})$, fix the particle-chart gauge of the even part by

$$\mathbb{I} - \hat{n}_{\hat{x}} = \lim_{\mu \rightarrow \infty} e^{-\mu \hat{n}_{\hat{x}}}, \quad D_\mu(\hat{x}) = \mathbb{I} - P_{\hat{x}} + e^{-\mu} P_{\hat{x}}. \quad (197)$$

At finite μ , this is an ordinary number-conserving Nambu transfer; its particle block is multiplied by $D_\mu(\hat{x})$. Therefore

$$T_{\text{eff}}^+ = \lim_{\mu \rightarrow \infty} T_p(\hat{W}\hat{V}) D_\mu(\hat{x}) = TS(\mathbb{I} - P_{\hat{x}}) = TS \left[\mathbb{I} - \frac{S^\dagger \chi \chi^\dagger S}{\chi^\dagger S S^\dagger \chi} \right]. \quad (198)$$

This is exactly the dressed-projector formula obtained from the Choi contraction calculation. It is a single-copy Nambu result. The apparent extra projector dressing comes from transporting the *regulated null vector* through the later nonunitary even segment before collapsing the endpoint sector.

The loss formula follows by the same argument in the hole chart. Start from

$$\hat{k}_\varepsilon^-(\chi) = \hat{c}(\chi) + \varepsilon \hat{c}^\dagger(\chi). \quad (199)$$

In Nambu-covector form,

$$w_\varepsilon^-(\chi) = \begin{pmatrix} \chi^* \\ \varepsilon \chi \end{pmatrix} \mapsto \tilde{w}_\varepsilon^-(\chi; S) = \begin{pmatrix} (S^{-1}\chi)^* \\ \varepsilon S^\dagger \chi \end{pmatrix}. \quad (200)$$

Equivalently, conjugating through $\hat{W}$ gives

$$\hat{W} \hat{k}_\varepsilon^-(\chi) \hat{V} = [\hat{c}(y) + \varepsilon \hat{c}^\dagger(x)] \hat{W} \hat{V}, \quad y = S^{-1}\chi, \quad x = S^\dagger \chi. \quad (201)$$

The dominant exact operator is now the annihilator in the normalized mode $\hat{y} = y/\|y\|$, with

$$P_{\hat{y}} = \frac{S^{-1}\chi \chi^\dagger S^{-\dagger}}{\chi^\dagger S^{-\dagger} S^{-1}\chi}. \quad (202)$$

Because loss is naturally reconstructed from the hole dictionary, the finite hole transfer is

$$T_{\text{eff},h}^- = (TS)^{-T} (\mathbb{I} - P_{\hat{y}}^T). \quad (203)$$

The homogeneous particle-side loss formula alone misses the endpoint source created by the exact annihilation; the hole formula Eq. (203) is the finite chart in which the singular sector has been removed.

Equations (198) and (203) also explain the common mistake. If one takes the endpoint limit at the insertion before composing with $\hat{W}$, one writes a projector P_χ in the wrong frame. The correct Nambu limit is

$$\begin{aligned} &\text{multiply/transport at finite regulator} \longrightarrow \text{identify the dominant null vector} \\ &\longrightarrow \text{take the rank-deficient endpoint.} \end{aligned} \quad (204)$$

For unitary S , $S^{-1}\chi = S^\dagger \chi$ and the distinction collapses; for nonunitary monitored evolution it is essential.

K. Neutral Choi Reference Versus Odd-Operator Regularization

There is an important distinction between two regularizations that are easy to conflate. The Pin/Nambu regulator above regularizes the odd trajectory amplitude $\hat{K}_\xi$ itself. The contraction rule in the Choi notes, however, regularizes the *channel* or superoperator

$$\mathcal{J}_K : \rho \mapsto \hat{K} \rho \hat{K}^\dagger \quad (205)$$

after vectorization. These are different one-body charts. They have the same exact endpoint after the correct scalar/postselection limit is taken, but their finite-regulator matrices do not have to look the same.

The clean Choi construction starts from a neutral maximally entangled reference. In a Majorana-neutral convention one may take, mode by mode,

$$|\Omega_M\rangle_{LR} = 2^{-N/2} \prod_{i=1}^N \left(1 + c_{L,i}^\dagger c_{R,i}^\dagger \right) |0\rangle, \quad (206)$$

for which a right-leg insertion can be moved to the left leg by a fixed particle-hole mirror, up to the usual graded signs:

$$c_{R,i}|\Omega_M\rangle_{LR} = -c_{L,i}^\dagger|\Omega_M\rangle_{LR}, \quad c_{R,i}^\dagger|\Omega_M\rangle_{LR} = c_{L,i}|\Omega_M\rangle_{LR}. \quad (207)$$

Thus, on the bare reference, left and right insertions are equivalent descriptions. After an operator has already acted, the same move becomes a graded transpose/mirror of the operator order; it does not permit one to commute a gain through later nonunitary factors.

For comparison with the Choi-contraction notes, one may also vectorize the channel $\mathcal{J}_K$. Introduce ket and bra copies, labelled $+$ and $-$, so that a gain jump is represented by the even bilinear

$$J_{\text{gain}}[\chi] = \hat{\chi}_+^\dagger \hat{\chi}_-, \quad J_{\text{loss}}[\chi] = \hat{\chi}_-^\dagger \hat{\chi}_+, \quad (208)$$

up to the fixed vectorization sign convention. These generators conserve the doubled charge $N_+ + N_-$. Since $J_{\text{gain}}^2 = J_{\text{loss}}^2 = 0$ on a fixed mode, their finite-regulator Gaussian families are unipotent rather than Pin reflections:

$$e^{\lambda J_{\text{gain}}[\chi]} : \quad \mathbb{G}_\lambda(\chi) = \mathbb{I} + \lambda |-\rangle\langle+| \otimes P_\chi, \quad (209)$$

$$e^{\mu J_{\text{loss}}[\chi]} : \quad \mathbb{L}_\mu(\chi) = \mathbb{I} + \mu |+\rangle\langle-| \otimes P_\chi, \quad (210)$$

where $P_\chi = \chi\chi^\dagger$. At every finite λ, μ these compose by literal ordered matrix multiplication in the doubled one-particle space. The exact jump is the projective endpoint

$$J_{\text{gain}}[\chi] = \lim_{\lambda \rightarrow \infty} \lambda^{-1} e^{\lambda J_{\text{gain}}[\chi]}, \quad J_{\text{loss}}[\chi] = \lim_{\mu \rightarrow \infty} \mu^{-1} e^{\mu J_{\text{loss}}[\chi]}. \quad (211)$$

This gives a second chart for the same exact endpoint and reproduces the composition rule obtained by explicit Choi contraction. It is not the same finite matrix family as $\hat{\chi}^\dagger + \varepsilon \hat{\chi}$ or $\hat{\chi} + \varepsilon \hat{\chi}^\dagger$.

For full-rank even Choi states with particle transfer matrices T and S , contracting the right leg of the first with the left leg of the second gives

$$R = TS. \quad (212)$$

The same statement holds on the doubled $+/-$ space with T, S replaced by $\mathbb{T}, \mathbb{S}$. For an exact gain, the isolated particle chart becomes rank deficient, $T \mapsto TQ$ with $Q = \mathbb{I} - P_\chi$, but a subsequent Choi contraction with an even piece S does not simply keep the undressed projector P_χ . The contraction derivation gives the dressed finite particle transfer

$$R_{\text{gain}} = TS(\mathbb{I} - P_{S^\dagger \chi}), \quad P_{S^\dagger \chi} = \frac{S^\dagger P_\chi S}{\chi^\dagger S S^\dagger \chi}. \quad (213)$$

Thus the gain constraint is transported contravariantly through the later factor S . The corresponding loss formula is most naturally read in the hole chart:

$$R_{\text{loss},h} = (TS)^{-T}(\mathbb{I} - P_{S^{-1}\chi}^T), \quad P_{S^{-1}\chi} = \frac{S^{-1}P_\chi S^{-\dagger}}{\chi^\dagger S^{-\dagger} S^{-1}\chi}. \quad (214)$$

Writing only the homogeneous particle-side expression for loss misses the source term carried by the emptied endpoint sector. This is precisely the dictionary obstruction: gain is reconstructed cleanly in the particle chart, loss in the hole chart, while mixed gain/loss composites can obstruct both fixed charts and must be handled in a graph/covariance chart. The doubled transfer chart below is one stable implementation of that contraction, not an additional assumption needed to derive the dressed projector.

The same-mode mixed product illustrates the relation to projective measurements. In the doubled chart,

$$\mathbb{G}_\lambda(\chi)\mathbb{L}_\mu(\chi) = \begin{pmatrix} \mathbb{I} & \mu P_\chi \\ \lambda P_\chi & \mathbb{I} + \lambda \mu P_\chi \end{pmatrix}_{+/-}. \quad (215)$$

Gauss decomposition of this block matrix shows that, as $\lambda, \mu \rightarrow \infty$, the unipotent factors trivialize and the Levi block degenerates to $Q = \mathbb{I} - P_\chi$ on the appropriate copy. This is the Choi-transfer version of

$$\hat{\chi} \hat{\chi}^\dagger = \mathbb{I} - \hat{n}_\chi, \quad \hat{\chi}^\dagger \hat{\chi} = \hat{n}_\chi.$$

Therefore the apparent discrepancy is only a discrepancy if the two regularizations are identified. The single-copy Nambu regulator already gives the dressed projector after the graph/covariance limit. The neutral/doubled Choi regulator is kept here only as a comparison chart in which channel contraction is also literal matrix multiplication.

For cumulative operator spectroscopy, the clean representation is the Choi state of the realized branch. Let $|V\rangle\rangle$ be the Choi vector of an even Gaussian operator V , and let χ_R denote the right-leg mode. The odd gain/loss amplitudes are represented by

$$|K_+\rangle\rangle = \hat{\chi}_R^\dagger |V\rangle\rangle, \quad |K_-\rangle\rangle = \hat{\chi}_R |V\rangle\rangle. \quad (216)$$

These are still pure Gaussian Choi states when their amplitudes are nonzero. The complication is therefore not a failure of Gaussianity or purity; it is a failure of the full-rank even particle-transfer chart.

Equivalently, exact right-leg gain/loss can be treated directly as occupation constraints on the Choi covariance. Let $\mathcal{C}$ be the particle correlation matrix of a normalized Choi Gaussian state on the doubled left/right one-particle space, and let X_R be a matrix whose columns are right-leg wavefunctions to be constrained, embedded in the doubled space with zero support on the left leg. The columns need not be orthonormal. Put

$$S_X = X_R^\dagger X_R, \quad P_X = X_R S_X^{-1} X_R^\dagger \quad (217)$$

when X_R has full column rank; if not, first remove the linearly dependent columns. A simultaneous gain constraint means that every mode in $\text{span } X_R$ is filled. Its raw ordered-operator norm is

$$p_{+X}^{\text{raw}} = \det[X_R^\dagger (\mathbb{I} - \mathcal{C}) X_R]. \quad (218)$$

If instead one wants the basis-independent occupation constraint associated only with the subspace P_X , the normalized probability is

$$p_{+X} = \frac{\det[X_R^\dagger (\mathbb{I} - \mathcal{C}) X_R]}{\det S_X}. \quad (219)$$

The conditional Choi covariance is independent of the chosen basis for the constraint subspace:

$$\mathcal{C}_{+X} = \mathcal{C} + (\mathbb{I} - \mathcal{C}) X_R [X_R^\dagger (\mathbb{I} - \mathcal{C}) X_R]^{-1} X_R^\dagger (\mathbb{I} - \mathcal{C}). \quad (220)$$

It satisfies

$$\mathcal{C}_{+X} P_X = P_X, \quad \Sigma_{+X} P_X = +P_X, \quad \Sigma_{LR,+X} P_X = 0, \quad (221)$$

where $\Sigma = 2\mathcal{C} - \mathbb{I}$. Thus the right-leg constrained subspace is an occupied endpoint sector.

The loss constraint is the particle-hole dual. The raw and normalized weights are

$$p_{-X}^{\text{raw}} = \det[X_R^\dagger \mathcal{C} X_R], \quad (222)$$

$$p_{-X} = \frac{\det[X_R^\dagger \mathcal{C} X_R]}{\det S_X}, \quad (223)$$

and, when the occupied Gram matrix is nonsingular,

$$\mathcal{C}_{-X} = \mathcal{C} - \mathcal{C} X_R [X_R^\dagger \mathcal{C} X_R]^{-1} X_R^\dagger \mathcal{C}. \quad (224)$$

Now

$$\mathcal{C}_{-X} P_X = 0, \quad \Sigma_{-X} P_X = -P_X, \quad \Sigma_{LR,-X} P_X = 0, \quad (225)$$

so the right-leg constrained subspace is an empty endpoint sector. Equations (220) and (224) are often the cleanest exact way to handle gain and loss in numerics: impose a right-leg occupation constraint on the Choi covariance, then read the finite transfer block only on the orthogonal complement of P_X .

Non-orthogonal wavefunctions cause no conceptual problem for same-type constraints. They only replace rank-one denominators by Gram determinants and Schur complements. What matters is the rank and conditioning of $X_R^\dagger (\mathbb{I} - \mathcal{C}) X_R$ for gains or $X_R^\dagger \mathcal{C} X_R$ for losses. A zero determinant means a blocked or redundant exact branch. It should be reported as endpoint/null-sector bookkeeping, not used as a finite transfer gap.

Mixed gain and loss constraints are different. A filled subspace and an empty subspace can be imposed simultaneously only when they are compatible; in particular, overlapping non-orthogonal directions cannot both be definitely filled and definitely empty. For example $\beta \hat{\alpha}^\dagger = \langle \beta | \alpha \rangle - \hat{\alpha}^\dagger \beta$, so changing the order changes the scalar contraction and the residual operator. The correct Choi-covariance procedure for a mixed non-commuting word is therefore the ordered

Pfaffian/Schur conditioning of Sec. VIB, applied on the doubled Choi state with all operators embedded on the appropriate leg. Only after that ordered conditioning has been performed does one identify endpoint multiplicity and the remaining finite Choi transfer spectrum.

To see this in the smallest possible example, take a single mode and $V = \mathbb{I}$. Both right-leg creation and right-leg loss have vanishing effective bridge between left and right particle sectors,

$$T_{\text{eff}} = 0. \quad (226)$$

Nevertheless they produce different Choi states:

$$\hat{\chi}_R^\dagger |\Omega\rangle \propto |1_L 1_R\rangle, \quad \hat{\chi}_R |\Omega\rangle \propto |0_L 0_R\rangle. \quad (227)$$

The singular matrix $T_{\text{eff}} = 0$ alone does not know whether the disconnected null sector is filled or empty.

For an isolated right-leg insertion after an even operator with particle transfer T_p , the rank-deficient particle-chart propagation part is

$$T_p \mapsto T_p(\mathbb{I} - P). \quad (228)$$

This is not the full Choi-contraction composition rule once later operators are present. If the constrained Choi state is contracted with a subsequent even transfer S , the projector is dressed as in Eqs. (213) and (214); gain transports χ by $S^\dagger$, while loss transports it by S^{-1} and should be reconstructed in the hole chart. The complementary hole chart obeys $T_h = (T_p^\dagger)^{-1}$ before the singular limit. Hence an exact particle-dark direction is a pole in the hole description, and an exact hole-dark direction is a pole in the particle description. The finite-sector spectrum must therefore be supplemented by endpoint data:

$$\{\lambda_{p,j}^{\text{finite}}\} \quad \text{plus} \quad (P_{\text{null}}, n_{\text{null}}, m_{\text{null}}), \quad (229)$$

where P_{null} is the projected direction, $n_{\text{null}} = 1$ for creation and $n_{\text{null}} = 0$ for annihilation, and m_{null} is the endpoint multiplicity.

In the Choi-covariance notation $\Sigma = 2C_{\text{Choi}} - \mathbb{I}$, the finite particle exponents can be read from the left-left block eigenvalues $a_j \in (-1, 1)$:

$$\lambda_{p,j}(t) = \frac{1}{2t} \log \left(\frac{1 - a_j(t)}{1 + a_j(t)} \right) = -\frac{1}{t} \operatorname{artanh} a_j(t), \quad a_j(t) \in \operatorname{spec} \Sigma_{LL}(t) \cap (-1, 1). \quad (230)$$

Endpoint sectors $a_j = \pm 1$ are formal infinite exponents in one of the particle/hole charts. A finite Lyapunov gap should therefore be defined as

$$\Delta_\lambda^{\text{finite}}(t) = \min_{a_j(t) \in (-1, 1)} \left| \frac{1}{t} \operatorname{artanh} a_j(t) \right|, \quad (231)$$

with the number of excluded endpoint eigenvalues reported separately. This is the gain/loss bookkeeping required before comparing transfer gaps across protocols or sizes.

L. Operational Conclusion for Non-Commuting Gain/Loss Strings

The practical rule is the following. If the trajectory contains a block of gains and losses, first read the word in chronological order and form the running charge Q_j of Eq. (130). Cut at returns to the running minimum. Each neutral block with nonzero causal Gram determinant gives an honest finite graph transfer through Eq. (135); its scalar $\lambda = \pm \det \tilde{G}$ is recorded in the Born-weight cocycle. Each unbalanced block is endpoint orbitals plus one finite matrix, as in Eq. (132), not a matrix by itself.

For same-type blocks this reduces to ordinary occupation constraints. Gains use the empty Gram matrix $X^\dagger(\mathbb{I} - C)X$, update the covariance by Eq. (48), and record the rank of the filled constrained subspace as endpoint data. Losses use the occupied Gram matrix $Y^\dagger CY$, update by Eq. (54), and record the rank of the emptied constrained subspace. For mixed non-commuting blocks, do not first combine all gains and all losses. Either apply the one-site gain/loss updates sequentially in chronological order, or build the ordered Pfaffian problem Eqs. (58)–(63); these are the same Wick calculation written in different forms.

In a fixed particle or hole transfer chart, the finite transfer matrix is therefore a reduced object obtained after the ordered constraints have been resolved. Its endpoint multiplicity is a rank computed from the ordered constrained

subspace. It is not, in general, the number of gain and loss events. Mixed gain/loss pairs can annihilate into projectors, produce scalar contractions through Eq. (56), or leave genuine charge-changing exterior/interior maps. The many-body spectrum must be built from the resulting ordered Fock-space map.

In a moving Majorana chart, the equivalent prescription is to replace each odd insertion by its regularized Pin representative, multiply all associated $O(2N, \mathbb{C})$ matrices in chronological order, and accumulate the projective scalar normalization separately. The singular limit $\varepsilon \rightarrow 0$ must be taken only after the ordered product and scalar weight have been formed. The matrix product before the limit is literally ordinary matrix multiplication; the limiting finite object is the graph/covariance block in Eq. (136), with endpoint/null sectors retained separately. This preserves the correct Born probability and prevents the divergent gauge directions of the regularized chart from being mistaken for finite CFT Lyapunov gaps.

VII. TANGENT COCYCLES FOR COVARIANCE DYNAMICS

The state covariance evolves by a branch-dependent nonlinear map

$$G_{t+1} = F_{\alpha_t}(G_t), \quad \alpha_t \sim p_t(\cdot | G_t). \quad (232)$$

For a Schur-complement Gaussian branch, a convenient form is

$$F_\alpha(G) = C_\alpha - B_\alpha^\dagger (A_\alpha - G^{-1})^{-1} B_\alpha. \quad (233)$$

Differentiating gives the tangent map. Let

$$M_\alpha(G) = A_\alpha - G^{-1}. \quad (234)$$

Since

$$\delta G^{-1} = -G^{-1} \delta G G^{-1}, \quad \delta M_\alpha = G^{-1} \delta G G^{-1}, \quad (235)$$

and $\delta M^{-1} = -M^{-1}(\delta M)M^{-1}$, one obtains

$$DF_\alpha(G)[\delta G] = B_\alpha^\dagger M_\alpha(G)^{-1} G^{-1} \delta G G^{-1} M_\alpha(G)^{-1} B_\alpha. \quad (236)$$

Using

$$M_\alpha(G)^{-1} G^{-1} = (GA_\alpha - \mathbb{I})^{-1}, \quad G^{-1} M_\alpha(G)^{-1} = (A_\alpha G - \mathbb{I})^{-1}, \quad (237)$$

define

$$T_\alpha(G) = (A_\alpha G - \mathbb{I})^{-1} B_\alpha. \quad (238)$$

When the Hermiticity relations of the covariance contraction identify the left factor with the adjoint of the right factor,

$$\delta G_{t+1} = T_{\alpha_t}(G_t)^\dagger \delta G_t T_{\alpha_t}(G_t). \quad (239)$$

Along a realized trajectory set $T_t = T_{\alpha_t}(G_{t-1})$. The chronological product convention used by Eq. (239) is

$$\tilde{T}_n = T_1 T_2 \cdots T_n, \quad \delta G_n = \tilde{T}_n^\dagger \delta G_0 \tilde{T}_n. \quad (240)$$

The single-particle tangent Lyapunov exponents are

$$\lambda_i^{\text{tan}} = \lim_{n \rightarrow \infty} \frac{1}{n} \log \sigma_i(\tilde{T}_n), \quad (241)$$

assuming the stationary ergodic hypotheses of the multiplicative ergodic theorem [5, 6]. These are quenched singular-value growth rates along sampled trajectories. In general,

$$\mathbb{E}[T_\alpha(G)^\dagger \delta G T_\alpha(G) | G] \neq \bar{T}(G)^\dagger \delta G \bar{T}(G), \quad \bar{T}(G) = \mathbb{E}[T_\alpha(G) | G]. \quad (242)$$

Thus the tangent Lyapunov spectrum is not the spectrum of an outcome-averaged transfer matrix.

VIII. TRAJECTORY FREE ENERGY AND CFT FINITE-SIZE DATA

For a monitored circuit on a cylinder of circumference L and time length t , define the Born-weighted trajectory free energy

$$F(L, t) = - \sum_{\xi} p_{\xi} \log p_{\xi} = -\mathbb{E}_{\xi \sim p} \log p_{\xi}. \quad (243)$$

In a Gaussian circuit, each p_{ξ} is computed by multiplying the determinant branch normalizations derived above. If the intrinsic space-time anisotropy is α , define

$$f_0(L) = \lim_{t \gg L} \frac{F(L, t)}{\alpha L t}. \quad (244)$$

At a conformally invariant non-unitary fixed point, the cylinder free-energy correction has the universal form [2, 3]

$$f_0(L) = f_0(\infty) - \frac{\pi c_{\text{eff}}}{6L^2} + \dots. \quad (245)$$

Here c_{eff} is the replica effective central charge relevant to the trajectory ensemble, not automatically the entanglement central charge of a prepared edge state.

A. Why Transfer Eigenvalues Are Free Energies

The transfer-matrix statement is just the row decomposition of a two-dimensional statistical mechanics sum. For a fixed circumference L , collect all local Boltzmann weights, Born weights, or trajectory amplitudes in one time strip into a linear operator $\mathcal{T}_L$ acting on the Hilbert space of boundary data around the cylinder. Gluing two neighboring strips means summing over their shared boundary variables, which is matrix multiplication. Gluing t strips and finally closing the time direction gives

$$Z(L, t) = \text{Tr}_{\text{bdry}} \mathcal{T}_L^t. \quad (246)$$

If $\mathcal{T}_L$ is diagonalizable with eigenvalues $\Lambda_A(L)$, this is

$$Z(L, t) = \sum_A \Lambda_A(L)^t. \quad (247)$$

For a non-Hermitian or trajectory-random problem one often computes singular values or Lyapunov exponents of products instead of ordinary eigenvalues, but the meaning is the same: the relevant number is the logarithmic growth or decay rate per time strip. Projecting onto, or inserting an operator that selects, a sector A gives

$$Z_A(L, t) \sim \Lambda_A(L)^t, \quad F_A(L, t) \equiv -\log Z_A(L, t) = -t \log \Lambda_A(L) + O(1), \quad (248)$$

where $\log \Lambda_A$ should be read as $\log |\Lambda_A|$ when phases do not affect the decay rate. Dividing by the cylinder area $\alpha L t$ therefore gives

$$f_A(L) = -\frac{\log |\Lambda_A(L)|}{\alpha L}, \quad f_A(L) - f_0(L) = \frac{-\log |\Lambda_A(L)| + \log |\Lambda_0(L)|}{\alpha L}. \quad (249)$$

This is the precise sense in which subleading transfer eigenvalues, singular values, or quenched Lyapunov exponents are “free-energy gaps.” The Lyapunov calculation is an efficient numerical route to the logarithms in Eq. (249); it is not an additional CFT postulate.

B. Replicas Are Not a Numerical Requirement

The numerical estimators in this note do not require introducing explicit replica degrees of freedom. A sampled trajectory directly supplies $-\log p_{\xi}$, and the typical gap is obtained by taking a logarithm of the trajectory-resolved transfer growth before averaging over ξ . Replicas are instead an analytic device: they rewrite quenched averages and trajectory moments as limits or continuations of integer-replica statistical mechanics models. The notation c_{eff}

refers to the effective central charge of that non-unitary trajectory ensemble. It should not be identified with the entanglement central charge $c = 1$ of a prepared chiral free-fermion edge unless a separate argument relates those two conformal theories.

Scaling dimensions are extracted from many-body transfer gaps, not directly from an arbitrary one-body Lyapunov spectrum. Let $\widehat{\mathcal{T}}_{\xi}(t)$ denote the unnormalized trajectory transfer operator in the sector whose correlation function is being probed, and let $S_A[\widehat{\mathcal{T}}_{\xi}(t)]$ be its many-body singular values ordered by decreasing typical growth rate. Define the quenched many-body gaps by

$$\mu_A(L) = - \lim_{t \rightarrow \infty} \frac{1}{t} \mathbb{E}_{\xi} \log \frac{S_A[\widehat{\mathcal{T}}_{\xi}(t)]}{S_0[\widehat{\mathcal{T}}_{\xi}(t)]}, \quad \mu_A(L) \geq 0. \quad (250)$$

The associated generalized free-energy gap is

$$\Delta f_A(L) = \frac{\mu_A(L)}{\alpha L}. \quad (251)$$

For low-lying operators of the effective CFT,

$$\Delta f_A(L) = \frac{2\pi x_A^{\text{typ}}}{L^2} + \dots, \quad \mu_A(L) = \frac{2\pi \alpha x_A^{\text{typ}}}{L} + \dots. \quad (252)$$

The superscript “typ” emphasizes that the logarithm is taken before the trajectory average.

For free fermions, Eq. (250) can often be reduced to one-body data, but only after reconstructing the exterior-power or Fock-space tower of Sec. V A. If the realized even Gaussian transfer object has finite one-body excitation rapidities $\epsilon_a(L)$, with endpoint/null sectors removed as in Sec. VI, then many-body gaps in a fixed charge/parity sector are additive combinations

$$\mu_I(L) = \sum_a n_a(I) \epsilon_a(L), \quad n_a(I) \in \mathbb{Z}_{\geq 0}, \quad (253)$$

up to the scalar normalization common to all states in the sector. The smallest positive one-body rapidity equals the first many-body gap only when the one-body list has already been measured relative to the dominant many-body configuration and the corresponding single-quasiparticle excitation is allowed by the sector and by the observable being measured. Otherwise the first CFT gap may be a difference of many-body sums, a sum of several one-body excitation rapidities, or a gap in a different particle/hole chart.

The tangent cocycle of Sec. VII is also first obtained as a one-body cocycle T_{ξ} . Its associated Gaussian many-body tangent transfer spectrum is again the additive Fock-space lift of the one-body Lyapunov data. Thus the raw λ_a^{tan} are not the final many-body gaps; the gaps are sums

$$\mu_I^{\text{tan}}(L) = \sum_a n_a(I) \epsilon_a^{\text{tan}}(L), \quad \epsilon_a^{\text{tan}}(L) \equiv |\lambda_a^{\text{tan}}(L) - \lambda_0^{\text{tan}}(L)|_{\text{finite}}, \quad (254)$$

with the sector I fixed by the perturbation or correlator being tested. Vectorizing δG is a useful way to implement the linear covariance map, but it does not by itself choose the many-body exterior-power sector or the excitation pattern $n_a(I)$.

Average correlations need not share the same exponent. If a trajectory correlator satisfies

$$G_{\xi}(t) \sim \exp\left[-\frac{2\pi t}{L} x_{\text{typ}}\right] \quad (255)$$

typically, its moments may instead scale as

$$\mathbb{E}[G_{\xi}(t)^n] \sim \exp\left[-\frac{2\pi t}{L} x(n)\right]. \quad (256)$$

Writing $Y = -\log G_{\xi}(t)$, multifractal scaling is the statement that the distribution of Y has the large-deviation form [7]

$$P_t(Y) \sim \left(\frac{2\pi \alpha t}{L}\right)^{-1/2} \exp\left[-\frac{2\pi \alpha t}{L} H\left(\frac{Y}{2\pi \alpha t/L}\right)\right], \quad (257)$$

with a nontrivial scaling function H . Expanding cumulants gives, for small n ,

$$x(n) = n x^{(1)} + \frac{n^2}{2} x^{(2)} + \dots, \quad x^{(1)} = x_{\text{typ}}. \quad (258)$$

The operational lesson is that a Lyapunov gap, an average correlator, and a typical correlator may be three different finite-size measurements of related but inequivalent universal data.

IX. MECHANICAL CPU EXTRACTION PLAN

The numerical extraction must be written so that every accumulated number has a named transfer object. For the domain-wall runs below the cylinder circumference is

$$L \equiv N_y, \quad (259)$$

and the anisotropy is reported with the default convention $\alpha = 1$ unless an independent calibration is supplied. The production CPU driver samples the physical Born circuit only:

$$\text{postselect} = \text{False}, \quad \text{postselect_probability} = 0. \quad (260)$$

Forced postselection is useful as a diagnostic trajectory, but it does not define the Born ensemble weight p_{ξ} and is therefore excluded from the c_{eff} estimator.

For sample s , size L , and completed cycle t , the CPU observer records the site-branch probabilities $p_{s,\ell}$ generated by the canonical covariance update. These include the Born probability of the measured occupation outcome and, when a stochastic correction is used, the probability of the sampled gain/loss correction. Deterministic perfect correction contributes unit probability. The accumulated trajectory action is

$$Y_s(L, t) = -\log p_{\xi_s(0:t)} = -\sum_{\ell \in [0, t)} \log p_{s,\ell}. \quad (261)$$

The free-energy density estimator is then

$$f_0(L, t) = \frac{1}{\alpha L t} \frac{1}{N_{\text{samp}}(L)} \sum_{s=1}^{N_{\text{samp}}(L)} Y_s(L, t). \quad (262)$$

After discarding early-time data and choosing a plateau time window, the cylinder fit is

$$f_0(L) = f_{\infty} - \frac{\pi c_{\text{eff}}}{6L^2} + O(L^{-4}). \quad (263)$$

Thus a linear fit of $f_0(L)$ against $1/L^2$ with slope m_0 gives

$$c_{\text{eff}} = -\frac{6m_0}{\pi}. \quad (264)$$

In the same physical trajectories the code accumulates two one-body Lyapunov objects and then builds the corresponding additive many-body gaps. The first is the covariance tangent cocycle,

$$\delta G_{t+1} = T_{\alpha_t}(G_t)^{\dagger} \delta G_t T_{\alpha_t}(G_t), \quad (265)$$

stabilized by QR decomposition after each cycle. The resulting exponents are called $\lambda_{j,s}^{\text{tan}}(L, t)$. The second is the Choi/operator covariance transfer object with the endpoint bookkeeping of Sec. VI; diagonalizing the finite particle/hole block after removing exact gain/loss endpoints gives $\epsilon_{j,s}^{\text{Choi}}(L, t)$. These are one-body rapidities from which many-body tangent and Choi/Kraus gaps must both be assembled additively, as in Eqs. (254) and (253). The analysis therefore reports

$$\delta \ell_{\text{tan}}^{(1)} \quad \epsilon_{\text{Choi}}^{(1)} \quad x_{\text{typ}}^{\text{tan/Fock}} \quad x_{\text{typ}}^{\text{Choi/Fock}}, \quad (266)$$

where the first two are raw one-body ingredients and the last two are CFT scaling-dimension estimates only after choosing allowed many-body Fock excitations.

For either one-body route $r \in \{\text{tan}, \text{Choi}\}$, the finite single-particle gap recorded by the CPU code is

$$\delta \ell_k^r(L, t) = \text{median}_s [\text{kth smallest}_j |\ell_{j,s}^r(L, t)|]_{\text{finite}}, \quad (267)$$

with $\ell^{\text{tan}} = \lambda^{\text{tan}}$ and $\ell^{\text{Choi}} = \epsilon^{\text{Choi}}$. Here the Choi rapidities are obtained from the finite covariance eigenvalues by the particle/hole chart convention, while the tangent exponents are obtained from QR-stabilized tangent-frame growth. The finite bracket means that zero/floor directions, endpoint directions, nonfinite values, and censored Choi trajectories are excluded before ordering. Their multiplicities are retained in the output table.

For each route the next mechanical step is to build the allowed many-body gaps

$$\mu_A^r(L, t) = \sum_j n_j^r(A) \delta \ell_j^r(L, t), \quad r \in \{\text{tan}, \text{Choi}\}, \quad (268)$$

with occupation pattern A fixed by the operator sector being tested. Only these μ_A^r , not the raw one-body list by itself, enter the CFT fit

$$\frac{\mu_A^r(L, t)}{\alpha L} = \frac{2\pi x_A^{\text{typ}}}{L^2} + O(L^{-3}). \quad (269)$$

The often-used estimator

$$x_{\text{one body}}^r(L, t) = \frac{L \delta \ell_1^r(L, t)}{2\pi\alpha} \quad (270)$$

is therefore a CFT scaling dimension only under the additional assumption that the first allowed many-body excitation is exactly one finite quasiparticle in route r . The CPU script parameter `fock_rank` = R implements only the mechanical choice

$$n_j^r(A_R) = \begin{cases} 1, & 1 \leq j \leq R, \\ 0, & j > R, \end{cases} \quad \mu_{A_R}^r(L, t) = \sum_{j=1}^R \delta \ell_j^r(L, t), \quad (271)$$

after ordering the finite one-body gaps. This is a transparent low-lying Fock-tower convention, not a claim that the physical operator of interest necessarily occupies the first R finite quasiparticle modes. If the sector requires a pair, a fixed charge difference, or a particle-hole combination, Eq. (268) must be used instead. Agreement between the tangent/Fock and Choi/Fock dimensions is evidence that the same slow channel is being observed by two Gaussian transfer constructions, not a proof by notation.

Gain and loss introduce the most important mechanical caveat. In the Choi route, eigenvalues satisfying

$$||g_j| - 1| < \varepsilon_{\text{end}} \quad (272)$$

are endpoint sectors and are not used as finite rapidity gaps. In the tangent route, QR directions whose diagonal factors hit the numerical floor, or whose cycle exponents become nonfinite, are classified as null/divergent sectors. Both endpoint counts are physical data: they measure how many rank-one creation/removal resets have left the chosen particle or hole chart. They are reported beside the finite-sector fits, not thrown away silently.

The concrete CPU workflow is therefore: run the canonical `classA.U1FGTN.run_markov_circuit` entry point with serial samples, enable the trajectory-weight observer, tangent observer, and Choi observer, save per-size raw spectra and probability actions, bootstrap over samples, fit $f_0(L)$ against $1/L^2$ for c_{eff} , plot the tangent and Choi one-body spectra, and convert both to x_{typ} only through the specified additive many-body Fock excitation.

For the current domain-wall adaptive circuit, the conservative interpretation is therefore: the observed small transfer-frame gap is evidence for a near-marginal interface channel localized on the wall. It is not, by itself, a precision extraction of x_i or c_{eff} . Such an extraction requires a controlled sequence in L , endpoint bookkeeping for gain/loss, and a clear choice of operator sector.

X. CAVEATS AND NON-IDENTIFICATIONS

1. The Born-sampled trajectory ensemble is not the deterministic mean channel. For nonlinear observables O ,

$$\mathbb{E}_{\xi}[O(G_{\xi})] \neq O(\mathbb{E}_{\xi}[G_{\xi}]). \quad (273)$$

2. A quenched product of random transfer matrices is not an averaged transfer matrix. Equation (242) is the minimal algebraic obstruction.
3. The Choi/Kraus particle spectrum and the tangent-cocycle spectrum are different linearizations. The former can generate many-body Gaussian transfer gaps after Fock-space reconstruction; the latter is a covariance stability spectrum unless tied to a specific trajectory correlator. Agreement between their slow sectors is physical evidence, not an operator identity.

4. Gain and loss are not harmless details. Exact creation or annihilation branches move the transfer description onto a singular boundary where a finite matrix must be supplemented by null-sector occupation data.
5. The entanglement central charge $c = 1$ of a chiral free-fermion edge is not automatically the same as the non-unitary effective central charge c_{eff} obtained from trajectory free energies. The former is a property of the prepared state; the latter is a property of the trajectory ensemble viewed as a replicated statistical mechanics problem.

XI. SUMMARY

In a free-fermion monitored circuit, the trajectory partition function $Z_{\xi} = p_{\xi}$ becomes a determinant calculation, and the many-body trajectory singular spectrum can be reconstructed from additive one-body singular data whenever the branch remains in a regular even Gaussian chart. Measurements, gain, and loss keep the state Gaussian but introduce singular transfer directions. The correct spectroscopy is therefore not simply “diagonalize a matrix”: one must state which matrix, which particle or hole chart, which endpoint sectors have been removed, which Fock-sector excitation is allowed, and whether the result concerns a cumulative operator or the tangent dynamics of covariance perturbations. Once that bookkeeping is explicit, the finite-size CFT formulas for c_{eff} , scaling dimensions, and multifractal spectra can be applied as a controlled diagnostic rather than as a slogan.

-
- [1] A. Zabalo, M. J. Gullans, J. H. Wilson, R. Vasseur, A. W. W. Ludwig, S. Gopalakrishnan, D. A. Huse, and J. H. Pixley, arXiv:2107.03393 (2022), [arXiv:2107.03393 \[cond-mat.dis-nn\]](#).
 - [2] J. L. Cardy, Nucl. Phys. B **270**, 186 (1986).
 - [3] P. Di Francesco, P. Mathieu, and D. Senechal, *Conformal Field Theory* (Springer, New York, 1997).
 - [4] P. Calabrese and J. Cardy, J. Stat. Mech. **2004**, P06002 (2004).
 - [5] V. I. Oseledets, Trans. Moscow Math. Soc. **19**, 197 (1968).
 - [6] H. Furstenberg and H. Kesten, Ann. Math. Statist. **31**, 457 (1960).
 - [7] F. Evers and A. D. Mirlin, Rev. Mod. Phys. **80**, 1355 (2008).
 - [8] C.-M. Jian, B. Bauer, A. Keselman, and A. W. W. Ludwig, Phys. Rev. B **106**, 134206 (2022).
 - [9] O. Alberton, M. Buchhold, and S. Diehl, Phys. Rev. Lett. **126**, 170602 (2021).
 - [10] M. Fava, L. Piroli, T. Swann, D. Bernard, and A. Nahum, Phys. Rev. X **13**, 041045 (2023).
 - [11] I. Poboiko, P. Popperl, I. V. Gornyi, and A. D. Mirlin, Phys. Rev. X **13**, 041046 (2023).
 - [12] K. Chahine and M. Buchhold, Phys. Rev. B **110**, 054313 (2024).
 - [13] A. W. W. Ludwig, H. Schulz-Baldes, and G. Stolz, J. Stat. Phys. **152**, 275 (2013).
 - [14] Z. Xiao and K. Kawabata, arXiv:2412.06133 (2024), [arXiv:2412.06133](#).
 - [15] A. Bhuiyan, H. Pan, and C.-M. Jian, Phys. Rev. Res. **8**, 023147 (2026), [arXiv:2507.13437](#).